

Optimization

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Section 1

Definitions

Constrained choice

Optimization selects the *best feasible point*.

- ◆ The objective assigns a value to every feasible point.
- ◆ Constraints determine the feasible set.

Example: maximize a utility function over the budget set of product bundles.

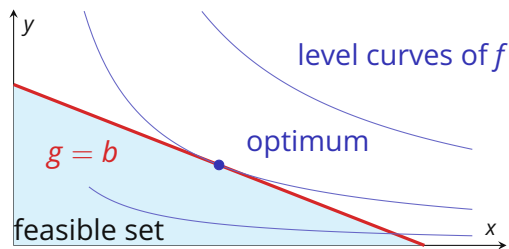


Figure 1 · Feasible set (shaded), level curves of the objective, and the best feasible point (optimum).

Optimization problems

General form. Given a feasible set $S \subseteq \mathbb{R}^n$ and an objective $f : S \rightarrow \mathbb{R}$,

$$\max_{x \in S} f(x).$$

Maximizer and maximum. In the set-builder and quantifier notation from Preliminaries,

$$\arg \max_{x \in S} f(x) := \{x^* \in S : \forall x \in S, f(x) \leq f(x^*)\}.$$

When this set is **nonempty**, the **maximum value** is $\max_{x \in S} f(x) = f(x^*)$ for every maximizer x^* .

Existence means $\arg \max_{x \in S} f(x) \neq \emptyset$; **uniqueness** means this set has one element.

Video: [Steve Brunton, *The anatomy of an optimization problem*](#).

Optimal value and optimal solution

Optimal value. Even with an empty argmax, the problem has an **optimal value**

$$v^* = \sup_{x \in S} f(x) \quad (\text{maximization}), \quad v^* = \inf_{x \in S} f(x) \quad (\text{minimization}),$$

where $\sup$ and $\inf$ are the least upper bound and the greatest lower bound of the value set $f(S) \subseteq \mathbb{R}$, with $\pm\infty$ allowed. An **optimal solution** is a feasible x^* with $f(x^*) = v^*$.

- ◆ When an optimal solution exists, $v^* = \max_{x \in S} f(x)$, the maximum value of the previous page.
- ◆ Comparative statics keeps this convention for the value function $V(\theta) = \sup_{x \in S(\theta)} f(x, \theta)$.

Every nonempty S has an optimal value; existence asks whether some feasible choice attains it, and a finite v^* is compatible with either answer.

Four questions in an optimization problem

Once the objective and feasible set are fixed, ask **four questions**.

Existence

Is $\arg \max_{x \in S} f(x)$ nonempty?

Globality and uniqueness

When is a local optimum global, and when is the optimizer unique?

Optimality conditions

Which conditions are necessary, and when are they sufficient?

Sensitivity

How do the optimal choice and optimized value change with parameters?

Compactness and continuity address existence; FOC, SOC, Lagrange and KKT give optimality conditions; **curvature** connects local and global conclusions; comparative statics and the envelope theorem study parameter changes.

Consumer and firm problems

The same template names the **choice variable**, the **feasible set**, and the **objective**.

Consumer choice

$$\max_{x \in \mathbb{R}_+^n} u(x) \quad \text{s.t.} \quad p \cdot x \leq w.$$

- ◆ x : consumption bundle.
- ◆ $p \cdot x \leq w$: budget feasibility.
- ◆ $u(x)$: objective to maximize.

Firm choice

$$\max_{q \geq 0} pq - c(q).$$

- ◆ q : output choice.
- ◆ $q \geq 0$: feasible production scale.
- ◆ $pq - c(q)$: profit objective.

From Convexity, the standard linear budget set is convex; when u represents the consumer's preferences, preference convexity is equivalent to quasiconcavity of u . Strict quasiconcavity makes an attained maximizer unique on the budget set.

Further reading: [Border, Preference and Demand Examples](#); [Border, Examples of Cost and Production Functions](#).

Data fitting and allocation

The same notation covers estimation and allocation; minimizing C is equivalent to maximizing $-C$.

Least squares fit

$$\min_{\beta \in \mathbb{R}^k} \|y - X\beta\|^2$$

- ◆ β : model coefficients.
- ◆ $\mathbb{R}^k$: unrestricted feasible set.
- ◆ $\|y - X\beta\|^2$: prediction error.

Resource allocation

$$\begin{aligned} \min_{a \in \mathbb{R}_+^m} C(a) \\ \text{s.t. } \sum_{i=1}^m a_i = R. \end{aligned}$$

- ◆ a_i : amount assigned to use i .
- ◆ $\sum_i a_i = R$: resource feasibility.
- ◆ $C(a)$: cost to minimize.

Local and global extrema

Let $x^* \in S$ and let $B_\epsilon^S(x^*) = S \cap B_\epsilon^{\mathbb{R}^n}(x^*)$, the open ball in S from Functions.

Global maximum $\forall x \in S, f(x) \leq f(x^*)$.

Local maximum $\exists \epsilon > 0 \forall x \in B_\epsilon^S(x^*), f(x) \leq f(x^*)$.

Global minimum $\forall x \in S, f(x) \geq f(x^*)$.

Local minimum $\exists \epsilon > 0 \forall x \in B_\epsilon^S(x^*), f(x) \geq f(x^*)$.

Global compares x^* with all of S ; local only with $B_\epsilon^S(x^*)$ for some $\epsilon > 0$. Since $B_\epsilon^S(x^*) \subseteq S$,

$$x^* \text{ global max} \Rightarrow x^* \text{ local max}, \quad x^* \text{ global min} \Rightarrow x^* \text{ local min}.$$

Hence a sufficient condition for a global extremum is sufficient for the corresponding local extremum, while a necessary condition for a local extremum is necessary for the corresponding global extremum. **The converses need not hold.**

Video: [Khan Academy, Multivariable maxima and minima.](#)

Local vs. global extrema

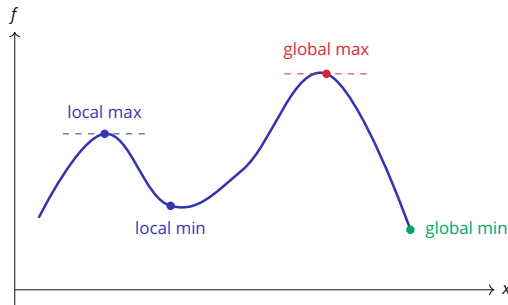


Figure 2 · Local and global extrema.

Local extrema compare nearby feasible points; a global extremum may occur at a boundary point.

A constrained maximum

Maximize $f(x)$ over $x \in [-1, 2]$.

- ◆ The unconstrained maximizer can lie outside the feasible set.
- ◆ The constrained maximizer can then lie on the boundary.

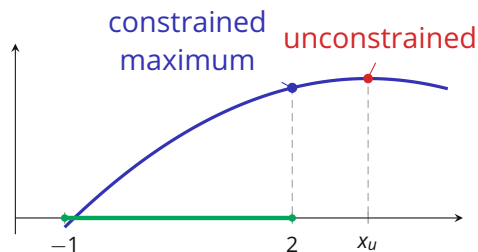


Figure 3 · The feasible interval is green.

The constrained maximum occurs at the boundary point $x = 2$.

Useful transformations

- ◆ Minimizers of f are maximizers of $-f$.
- ◆ If g is strictly increasing on the range of f , then composition preserves the optimizer set:

$$\arg \max_{x \in S} f(x) = \arg \max_{x \in S} (g \circ f)(x).$$

Log trick. For $f(x) = A \prod_i x_i^{\alpha_i} > 0$, $\ln f = \ln A + \sum_i \alpha_i \ln x_i$. Maximizing a positive likelihood $L(\theta)$ is likewise equivalent to maximizing $\ell(\theta) = \ln L(\theta)$.

This applies composition and monotonicity from Functions: $g \circ f$ preserves the ranking of feasible points when g is **strictly increasing**. The **optimizer set is unchanged**, but the numerical value of the objective changes.

Further reading: Boyd, Ch. 20, §§20.3.3–20.3.5.

Monotone transformations and optimizers

Let $I \subseteq \mathbb{R}$ contain every value of $f : S \rightarrow \mathbb{R}$, and let $g : I \rightarrow \mathbb{R}$.

- (a) For g strictly increasing, prove $\arg \max_{x \in S} f(x) = \arg \max_{x \in S} g(f(x))$.
- (b) Does the maximum value stay the same?
- (c) State and prove the version for g strictly decreasing.
- (d) Apply (a) to $f(x, y) = x^\alpha y^\beta$ on $S \subseteq \mathbb{R}_{++}^2$ with $g = \ln$.

Monotone transformations and optimizers

- (a) If $x^* \in \arg \max_S f$, then $f(x) \leq f(x^*)$ on S , so $g(f(x)) \leq g(f(x^*))$. Conversely, if x^* maximizes $g \circ f$ and some $x \in S$ had $f(x) > f(x^*)$, then $g(f(x)) > g(f(x^*))$, a contradiction.
- (b) The maximizers are the same points, and the value becomes $\max_S g \circ f = g(\max_S f)$.
- (c) $\arg \max_{x \in S} f(x) = \arg \min_{x \in S} g(f(x))$, by the same two steps with the inequalities reversed.
- (d) $\ln$ is strictly increasing on $(0, \infty)$, so the transformed objective is $\alpha \ln x + \beta \ln y$ with the same maximizers: the log trick.

Preliminaries applied: logic of an optimizer

Let $S \subseteq \mathbb{R}^n$, $f : S \rightarrow \mathbb{R}$, and $x^* \in S$. The definition of the previous pages says

$$x^* \in \arg \max_{x \in S} f(x) \iff \forall x \in S, f(x) \leq f(x^*).$$

- (a) Negate the statement on the right.
- (b) Write, with quantifiers, the statement that x^* is the unique maximizer.
- (c) Given that x^* is a maximizer, negate uniqueness.
- (d) Say why $\arg \max_{x \in S} f(x)$ is a set while $\max_{x \in S} f(x)$ is a number whenever a maximizer exists.

Preliminaries applied: logic of an optimizer

- (a) $\exists x \in S$ such that $f(x) > f(x^*)$.
- (b) $x^* \in S$ and $\forall x \in S \setminus \{x^*\}$, $f(x) < f(x^*)$; in set form, $\arg \max_{x \in S} f(x) = \{x^*\}$.
- (c) Uniqueness fails when $\exists x \in S \setminus \{x^*\}$ with $f(x) = f(x^*)$. The strict inequality of (b) fails at such an x , and $f(x) > f(x^*)$ is ruled out because x^* is a maximizer, so equality is the one remaining case.
- (d) Several feasible points can satisfy the same universal statement, and the definition collects all of them, so the argmax is a set. The maximum is the common value $f(x^*)$ at those points, one number.

Maximizers, maximum values, and local extrema

Let $f(x) = x^3 - 3x$ on $S = [-2, 2]$.

- (a) Find $\arg \max_{x \in S} f(x)$ and $\max_{x \in S} f(x)$.
- (b) Classify $x = -1$, $x = 1$, and $x = 2$ as local or global maxima or minima relative to S .
- (c) Explain why the answer to part (a) is a set, whereas the maximum is a number.

Maximizers, maximum values, and local extrema

Since $f'(x) = 3(x^2 - 1)$, the interior critical points are $x = \pm 1$. Evaluating them and the endpoints gives

$$f(-2) = -2, \quad f(-1) = 2, \quad f(1) = -2, \quad f(2) = 2.$$

Hence

$$\arg \max_{x \in S} f(x) = \{-1, 2\}, \quad \max_{x \in S} f(x) = 2.$$

The point -1 is a local and global maximum; 2 is a boundary local maximum and a global maximum. The point 1 is a local and global minimum. The $\arg\max$ records the choices attaining the value; the maximum records the attained value itself.

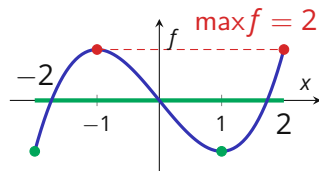


Figure 4 · $f(x) = x^3 - 3x$ on $[-2, 2]$: two maximizers, one maximum value.

Section 2

Existence, FOC & SOC

Does an optimization problem attain its optimum?

Here, existence means that at least one feasible choice attains the largest objective value:

$$\arg \max_{x \in S} f(x) \neq \emptyset.$$

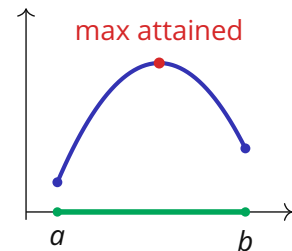
- ◆ Not every optimization problem has a maximizer: $\max_{x \in \mathbb{R}} x$ has none.
- ◆ From Functions: if $S \subseteq \mathbb{R}^n$ is **nonempty and compact** and $f : S \rightarrow \mathbb{R}$ is **continuous**, a maximizer $x^* \in S$ exists (Extreme Value Theorem, Weierstrass).
- ◆ Concavity serves a **different role**: it helps turn local conditions into global conclusions.

Logical order: first ask whether a solution exists; then derive optimality conditions and distinguish necessity from sufficiency; then use curvature, or compare the candidate values directly, to establish global optimality and uniqueness.

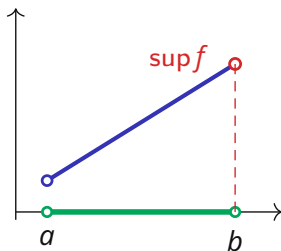
Further reading: Boyd, §29.42+ and §30.2; Carter, Ch. 6, pp. 601–602; §6.2, pp. 609–621.

Video: Mike, the Mathematician, *Continuous Functions on Compact Sets and the Extreme Value Theorem*.

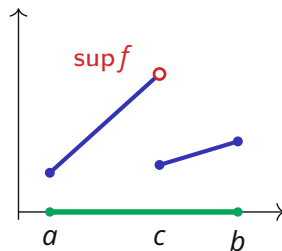
Attained maximum versus unattained supremum



closed interval,
continuous f : attained



open interval:
supremum at
the open end



jump at c :
supremum at
the missing limit

Figure 5 · The green segment is the feasible set, with a filled endpoint where it belongs to the set and a hollow one where it does not. Left: a compact feasible set and a continuous objective attain the maximum. Middle: an open endpoint removes the point where the supremum would be reached. Right: a jump does the same on a closed interval. Each hypothesis of the Extreme Value Theorem blocks one failure.

Consumer choice: why a maximizer exists

A standard linear budget set with strictly positive prices is compact

Let $p \gg 0$, $w > 0$, and $B(p, w) = \{x \in \mathbb{R}_+^n : p \cdot x \leq w\}$.

Closed. The map $x \mapsto p \cdot x$ is continuous, so from Functions $\{x \in \mathbb{R}^n : p \cdot x \leq w\}$, its preimage of the closed set $(-\infty, w]$, is closed; so is each $\{x : x_i \geq 0\}$. Their intersection $B(p, w)$ is closed, since its complement is a union of open sets.

Bounded. If $x \in B(p, w)$, then $0 \leq x_i \leq w/p_i$ for each i .

Compact. Closed and bounded in $\mathbb{R}^n$ is compact (Heine–Borel, Functions), so for every continuous $u : B(p, w) \rightarrow \mathbb{R}$ the Extreme Value Theorem gives $\arg \max_{x \in B(p, w)} u(x) \neq \emptyset$.

Linear Algebra supplies the inner product $p \cdot x$; Functions supplies the preimage, closed and bounded sets, compactness and the EVT; the argmax is the set defined on the first Definitions page.

Compactness versus convexity

Compactness can fail.

With $p \gg 0$, $w > 0$, the set $\{x \in \mathbb{R}_+^n : p \cdot x < w\}$ is bounded and misses the limit of $(1 - 1/k)(w/p_1)e_1$, so it is not closed.

At $p = (1, 0)$, $w = 1$, the set $\{x \in \mathbb{R}_+^2 : x_1 \leq 1\}$ is closed and unbounded.

Convexity can fail.

$\{x \in [0, 1]^2 : x_1 x_2 = 0\}$, the two axis segments, is compact, but the midpoint of $(1, 0)$ and $(0, 1)$ is outside it.

Linear inequalities always give convexity, and $x_1^2 + x_2^2 \leq 1$ shows a nonlinear constraint can too.

With $p \gg 0$, the weak inequality makes $B(p, w)$ closed, positive prices make it bounded, and linearity makes it convex. **Existence uses compactness with continuity; convexity is used later, for globality and uniqueness.**

Concavity and convexity

Convexity of the feasible set keeps every segment from a local candidate to another feasible point inside the feasible set.

Proposition. Let $S \subseteq \mathbb{R}^n$ be convex and let $f : S \rightarrow \mathbb{R}$.

- (i) If f is convex, every local minimizer is global. If f is strictly convex, f has **at most one** minimizer.
- (ii) If f is concave, every local maximizer is global. If f is strictly concave, f has **at most one** maximizer.

Further reading: MIT 6.253, *Convex Analysis and Optimization*, p. 59; Boyd and Vandenberghe, *Convex Optimization*, §4.2.

Video: Mohamed Omar, *Convex: Local vs Global Minimum*.

Existence, attainment, and uniqueness

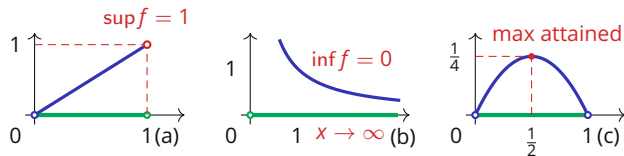
For (a) to (c), report the optimal value and the solution set, and say which hypothesis of the Extreme Value Theorem fails.

$$(a) \max_{x \in (0,1)} x, \quad (b) \min_{x \in \mathbb{R}_{++}} \frac{1}{x}, \quad (c) \max_{x \in (0,1)} x(1-x).$$

- (d) State what each pair of hypotheses implies: nonempty compact S with continuous f ; convex S with concave f ; convex S with strictly concave f . Then give a strictly concave f on a convex S whose argmax is empty.

Existence, attainment, and uniqueness: (a) to (c)

- (a) $v^* = \sup_{x \in (0,1)} x = 1$ with an empty argmax: for feasible x , the feasible point $(1+x)/2$ is larger. The feasible set fails compactness.
- (b) $v^* = \inf_{x \in \mathbb{R}_{++}} 1/x = 0$ with an empty argmin: $1/x > 0$ at every feasible x while $1/k \rightarrow 0$, and $1/(2x) < 1/x$. No feasible choice attains 0, so $\min_{x \in \mathbb{R}_{++}} 1/x$ is undefined. Against $\max_{x \in \mathbb{R}} x$, which has $v^* = +\infty$: there the value is infinite, here it is finite and unattained.
- (c) $x(1-x) = \frac{1}{4} - (x - \frac{1}{2})^2$, so $x^* = 1/2$ and $v^* = 1/4$. The same hypothesis fails as in (a), and a maximizer exists: the theorem gives existence and says nothing when its hypotheses fail.



Existence, attainment, and uniqueness: (d)

(d) Each pair of hypotheses gives one conclusion:

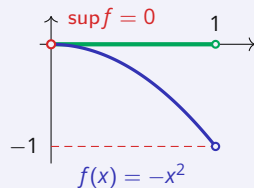
$S \neq \emptyset$ compact, f continuous $\Rightarrow \arg \max f \neq \emptyset$

S convex, f concave $\Rightarrow$ every local maximum is global

S convex, f strictly concave $\Rightarrow$ at most one maximizer

The third row bounds the number of maximizers by one.

Existence still comes from compactness: $f(x) = -x^2$ is strictly concave on the convex set $(0, 1)$, where $v^* = 0$ and the argmax is empty.



Stationarity

FONC. If f is **differentiable** at x^* and x^* is an **interior** local extremum, then

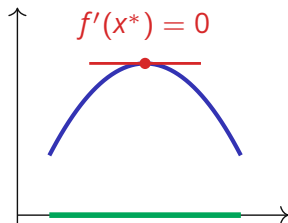
$$\nabla f(x^*) = 0.$$

1. Solve $\nabla f(x) = 0$ to find interior critical points.
2. Check the **boundary** and **non-differentiable** points.
3. Compare objective values when several feasible possibilities remain.

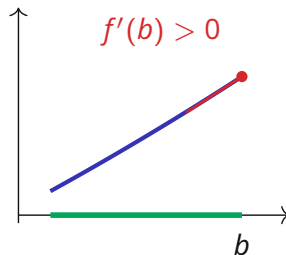
Reduce to one variable: fix i and set $\varphi_i(t) = f(x^* + te_i)$, with e_i the i th coordinate vector. Since x^* is interior, φ_i is defined on an interval around 0 and has a local extremum at $t = 0$, so $\varphi_i'(0) = 0$. The chain rule from Differentiation gives $\varphi_i'(0) = \nabla f(x^*)^\top e_i = \partial f(x^*)/\partial x_i$. Every partial derivative vanishes, so $\nabla f(x^*) = 0$. **Under the assumptions of the FONC, stationarity is necessary, not sufficient.**

Videos: [Arizona Math Camp 30, unconstrained optimization](#); Krista King, *Critical points*.

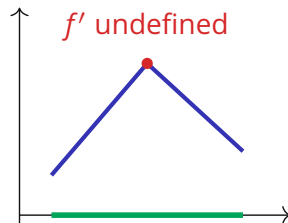
Three places a maximum can lie



interior: stationary



boundary: an endpoint



kink: a corner

Figure 6 · Interior maximum with a horizontal tangent; boundary maximum where the derivative is positive; kink where the derivative fails to exist. Solving $f'(x) = 0$ finds the first case only.

At an interior critical point

For f twice continuously differentiable near x^* , the second-order Taylor expansion at $x^* + h$ is

$$f(x^* + h) - f(x^*) = \nabla f(x^*)^\top h + \frac{1}{2} h^\top H_f(x^*) h + o(\|h\|^2).$$

At an interior critical point, $\nabla f(x^*) = 0$, so

$$f(x^* + h) - f(x^*) = \frac{1}{2} Q(h) + o(\|h\|^2), \quad Q(h) := h^\top H_f(x^*) h.$$

Local maximum. Necessary: $H(x^*) \preceq 0$.

Sufficient: $H(x^*) \prec 0$.

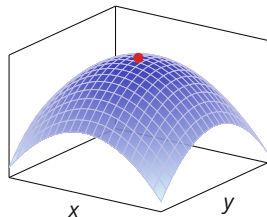
Local minimum. Necessary: $H(x^*) \succeq 0$.

Sufficient: $H(x^*) \succ 0$.

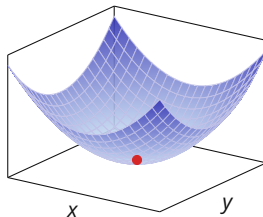
This combines Taylor expansion and the Hessian from Differentiation with the quadratic-form sign test from Convexity. Indefinite gives a saddle; **with a semidefinite Hessian the test is inconclusive.**

Videos: [Trefor Bazett, the second derivative test](#); MIT 18.02SC, [Second derivative test](#).

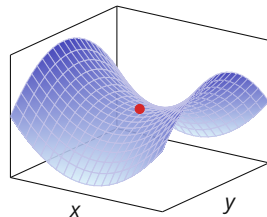
Hill, bowl and saddle



$f = -x^2 - y^2, H = \begin{bmatrix} -2 & 0 \\ 0 & -2 \end{bmatrix} \prec 0$: the surface falls away in every direction, a maximum.



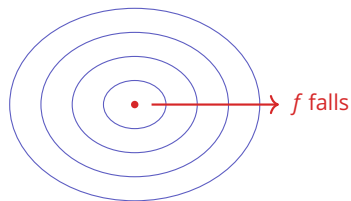
$f = x^2 + y^2, H = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \succ 0$: the surface rises in every direction, a minimum.



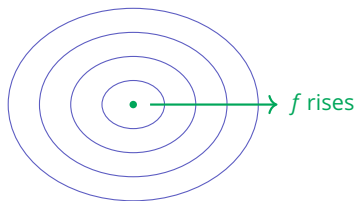
$f = x^2 - y^2, H = \begin{bmatrix} 2 & 0 \\ 0 & -2 \end{bmatrix}$ indefinite: it rises along $y = 0$ and falls along $x = 0$, a saddle.

Figure 7 · The three sign cases of the quadratic form $\frac{1}{2} h^T H h$ at a critical point, with the point marked in red. The next page shows the same three cases through their level curves, which is each surface seen from above.

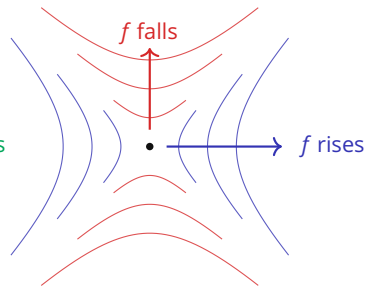
Level curves near a critical point



$H \prec 0$: local max
 f falls outward



$H \succ 0$: local min
 f rises outward



H indefinite: saddle
 f rises along one axis, falls along the other

Figure 8 · Level curves of f around a critical point. Negative definite Hessian: closed curves with f falling outward, a local maximum. Positive definite: f rising outward, a local minimum. Indefinite: hyperbolas, f rises along one axis and falls along the other, a saddle.

Local test versus global test

Suppose $S \subseteq \mathbb{R}^n$ is convex and f is twice continuously differentiable on an open set containing S . The same Hessian appears in both tests; the quantifier is what changes.

Convexity on S

$H_f(x) \succeq 0$ **for every** $x \in S \Rightarrow f$ is convex on S .

If $H_f(x) \succ 0$ **for every** $x \in S$, then f is strictly convex on S .

Concavity on S

$H_f(x) \preceq 0$ **for every** $x \in S \Rightarrow f$ is concave on S .

If $H_f(x) \prec 0$ **for every** $x \in S$, then f is strictly concave on S .

Convexity applied. Definiteness of $H_f(x^*)$ at the **single point** x^* gives only a **local** conclusion; a sign that holds for **every** $x \in S$ establishes convexity or concavity on S , so the proposition on concavity and convexity applies. A boundary optimum need not satisfy $\nabla f(x^*) = 0$, and Figure 9 shows a case where the sign of f'' changes inside S .

Further reading: Video: [Steve Brunton, Convexity 101](#). Boyd, Ch. 21.

One point test, one global test

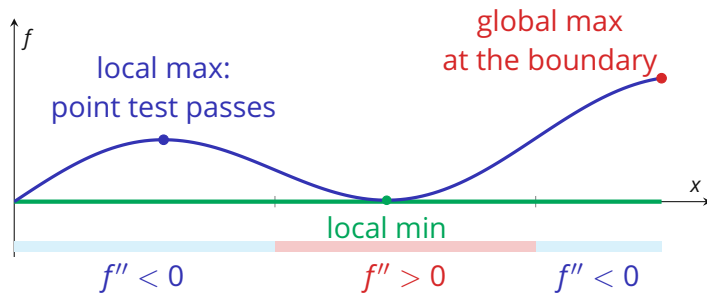


Figure 9 · At the local maximum $f'' < 0$, so the point test passes. The sign of f'' changes on S , so f is concave on only part of S and the local conclusion stays local: the global maximum lies on the boundary.

A quadratic objective

On $(x_1, x_2) \in [0, 4]^2$, let

$$f(x_1, x_2) = 2x_1x_2 - x_1^2 - 2x_2^2 + 3x_1 - 2x_2.$$

First-order conditions

$$\nabla f(x) = 0 \quad \Rightarrow \quad (x_1^*, x_2^*) = (2, \tfrac{1}{2}).$$

Second-order test

$$H_f = \begin{pmatrix} -2 & 2 \\ 2 & -4 \end{pmatrix}.$$

$$D_1 = -2 < 0 \text{ and } D_2 = 4 > 0, \text{ so } H_f \prec 0.$$

The constant Hessian is negative definite, so f is strictly concave on $[0, 4]^2$. Hence $(2, \frac{1}{2})$ is the unique global maximizer. Since f is quadratic, its second-order Taylor formula has no higher-order remainder.

Linear Algebra and Differentiation: least squares

From squared error to normal equations

Begin with the unrestricted problem

$$\min_{\beta \in \mathbb{R}^k} S(\beta), \quad S(\beta) = (y - X\beta)^\top (y - X\beta).$$

Expanding the objective gives

$$S(\beta) = y^\top y - 2\beta^\top X^\top y + \beta^\top X^\top X \beta.$$

Since $X^\top X$ is symmetric,

$$\nabla_\beta S(\beta) = -2X^\top y + 2X^\top X \beta.$$

The FOC is the system of normal equations

$$X^\top X \hat{\beta} = X^\top y.$$

Linear Algebra and Convexity: least squares

Solving the normal equations

If X has full column rank, then $X^\top X$ is positive definite and invertible. Hence

$$\hat{\beta} = (X^\top X)^{-1} X^\top y.$$

The Hessian is

$$H_S = 2X^\top X \succ 0,$$

so the critical point is the unique global minimizer. Full column rank and invertibility come from Linear Algebra; positive definiteness converts the Hessian calculation into strict convexity. The Equality constraints section adds the affine restriction $R\beta = r$ to this problem.

Least squares as a projection

The geometry of the normal equations

The normal equations $X^T(y - X\hat{\beta}) = 0$ say that the residual is orthogonal to every column of X .

Hence $X\hat{\beta}$ is the point of $\text{col}(X)$ nearest to y : minimizing $\|y - X\beta\|^2$ is projecting y onto the column space.

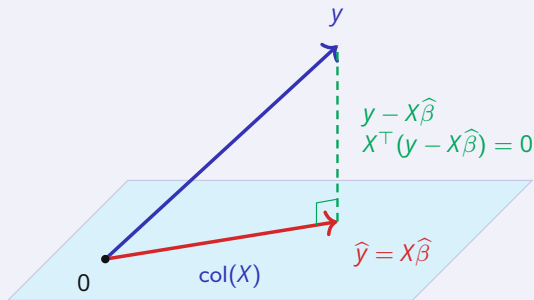


Figure 11 · The fitted vector is the orthogonal projection of y onto $\text{col}(X)$.

Video: MIT 18.06, Strang, *Projection Matrices and Least Squares*.

Linear Algebra applied: FOC as a linear system

Let

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}, \quad b = \begin{bmatrix} 5 \\ 4 \end{bmatrix}, \quad f(x) = b^\top x - \frac{1}{2} x^\top A x.$$

- (a) Show that the FOC is the linear system $Ax = b$.
- (b) Compute A^{-1} and the critical point.
- (c) Use the Hessian to determine whether the critical point is the unique global maximizer.

Linear Algebra applied: FOC as a linear system

Since A is symmetric,

$$\nabla f(x) = b - Ax, \quad \nabla f(x) = 0 \Rightarrow Ax = b.$$

Moreover,

$$\det A = 3, \quad A^{-1} = \frac{1}{3} \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}, \quad x^* = A^{-1}b = \begin{bmatrix} 2 \\ 1 \end{bmatrix}.$$

The Hessian is $-A$. The leading principal minors of A are $2 > 0$ and $3 > 0$, so $A \succ 0$, $-A \prec 0$, and f is strictly concave. Thus x^* is the unique global maximizer.

Necessary and sufficient conditions

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be twice differentiable and let x^* be an interior point. Use the implication definitions of necessary and sufficient conditions from Preliminaries. For the statement “ x^* is a strict local maximum,” classify each condition as necessary, sufficient, both, or neither:

$$(a) f'(x^*) = 0, \quad (b) f''(x^*) < 0,$$

$$(c) f'(x^*) = 0 \text{ and } f''(x^*) < 0.$$

Give a counterexample for every failed implication.

Necessary and sufficient conditions

- (a) Necessary but not sufficient. Fermat's theorem gives necessity; $f(x) = x^3$ at $x^* = 0$ shows that stationarity is not sufficient.
- (b) Neither. The function $f(x) = -x^4$ has a strict maximum at zero with $f''(0) = 0$. Conversely, $f(x) = x - x^2$ has $f''(0) < 0$, but zero is not a local maximum because $f'(0) = 1$.
- (c) Sufficient but not necessary. The usual derivative test using f'' gives sufficiency, while $f(x) = -x^4$ again shows failure of necessity.

Boundary points, saddles, and an inconclusive SOC

- (a) Maximize $f(x) = -(x - 3)^2$ on $[0, 2]$. Identify the role of the boundary.
- (b) Classify the critical point of $g(x, y) = x^2 - y^2$.
- (c) At $x = 0$, the second derivative of both $h(x) = x^4$ and $k(x) = -x^4$ is zero. Classify the point for each function and state what the SOC concludes.

Boundary points, saddles, and an inconclusive SOC

On $[0, 2]$, $f'(x) = 6 - 2x > 0$, so f is increasing and its maximum is $f(2) = -1$ at the boundary $x = 2$; there is no interior stationary point. For g , the only critical point is $(0, 0)$, and

$$H_g = \begin{bmatrix} 2 & 0 \\ 0 & -2 \end{bmatrix}$$

is indefinite, so $(0, 0)$ is a saddle. Finally, $h(x) \geq 0 = h(0)$, so 0 is a strict global minimum of h ; $k(x) \leq 0 = k(0)$, so 0 is a strict global maximum of k . In both cases the zero second derivative makes the SOC inconclusive.

Section 3

Equality constraints

The Lagrange multiplier condition

Problem. $\max_{x \in \mathbb{R}^n} f(x)$ subject to $g_j(x) = b_j, j = 1, \dots, m$. Write $g = (g_1, \dots, g_m)$, so the constraint is $g(x) = b$ and $Dg(x)$ is the $m \times n$ Jacobian with rows $\nabla g_j(x)^\top$.

Lagrange multiplier theorem. If x^* is a local optimum of the problem and $\text{rank } Dg(x^*) = m$, then for some $\lambda^* \in \mathbb{R}^m$

$$\nabla f(x^*) = Dg(x^*)^\top \lambda^* = \sum_{j=1}^m \lambda_j^* \nabla g_j(x^*).$$

At a regular constrained optimum, $\nabla f(x^*) \in \text{span}\{\nabla g_1(x^*), \dots, \nabla g_m(x^*)\}$: **necessary but not sufficient**, since a constrained minimum solves it too. Each λ_j^* **can have either sign**; $g_j \leq b_j$ forces $\lambda_j^* \geq 0$ on the KKT pages.

The rank hypothesis is a **constraint qualification**: the gradients $\nabla g_1(x^*), \dots, \nabla g_m(x^*)$ are **linearly independent**, for equality constraints the LICQ.

Why the Lagrange multiplier condition holds

Tangent space and the implicit function theorem

A feasible curve through x^* has velocity d with $Dg(x^*)d = 0$. When $\text{rank } Dg(x^*) = m$, the implicit function theorem of Differentiation gives the converse: every such d is the velocity of a feasible curve through x^* , so the first-order feasible directions form the tangent space $T(x^*) = \ker Dg(x^*)$.

Along such a curve $t \mapsto f(x(t))$ has a local maximum at $t = 0$, so its derivative vanishes:

$$\nabla f(x^*)^\top d = 0 \text{ for every } d \in T(x^*), \quad \text{that is} \quad \nabla f(x^*) \in (\ker Dg(x^*))^\perp.$$

From Linear Algebra $(\ker Dg(x^*))^\perp = \text{im } Dg(x^*)^\top$, the span of the rows of $Dg(x^*)$, which are $\nabla g_1(x^*), \dots, \nabla g_m(x^*)$. Hence

$$\nabla f(x^*) = \sum_{j=1}^m \lambda_j^* \nabla g_j(x^*) = Dg(x^*)^\top \lambda^*$$

for some $\lambda^* \in \mathbb{R}^m$. The rank hypothesis enters once, at the converse.

Video: [Arizona Math Camp 34, geometry of constrained optimization.](#)

When regularity fails

$$\max_x x \quad \text{s.t.} \quad g(x) = x^2 = 0.$$

The feasible set is $\{0\}$, so $x^* = 0$ is the unique global maximizer. At that point

$$f'(0) = 1, \quad g'(0) = 0, \quad \text{so} \quad \text{rank } Dg(0) = 0 < 1 = m,$$

and **the constraint qualification of the theorem fails**. No multiplier λ satisfies $f'(0) = \lambda g'(0)$, so the theorem's conclusion fails at this maximizer: it is **necessary only where a constraint qualification holds**.

Where the derivation breaks: $\ker Dg(0) = \mathbb{R}$, so the linearized feasible directions fill the whole line, while the feasible set is the single point 0 and no feasible curve moves at all. The rank condition is what makes $\ker Dg(x^*)$ describe the feasible set near x^* ; without it the linearization overstates the feasible directions, and the multiplier it would supply need not exist.

The Lagrangian

$$\mathcal{L}(x, \lambda) = f(x) + \sum_{j=1}^m \lambda_j (b_j - g_j(x)).$$

- ◆ $\nabla_x \mathcal{L}(x^*, \lambda^*) = 0$ is the multiplier condition $\nabla f(x^*) = \sum_j \lambda_j^* \nabla g_j(x^*)$ in one line, and $\nabla_\lambda \mathcal{L} = 0$ returns the constraints $g_j(x^*) = b_j$. Solving the two together is the standard route to a candidate (x^*, λ^*) .

Proposition. Let every g_j be affine and let (x^*, λ^*) satisfy the FOCs: x^* is feasible and $\nabla f(x^*) = \sum_j \lambda_j^* \nabla g_j(x^*)$. Then

f concave $\implies x^*$ is a global maximizer, f convex $\implies x^*$ is a global minimizer.

If f is strictly concave, or strictly convex, that optimizer is **unique**.

Boyd, Ch. 18. Videos: Trefor Bazett, *Lagrange Multipliers: Geometric Meaning and Example*; MIT 18.02SC, *Lagrange multipliers*.

A first Lagrange problem, solved in full

Problem and Lagrangian

$$\max_{x,y} 6x + 4y - x^2 - y^2 \quad \text{s.t.} \quad x + y = 4,$$

$$\mathcal{L} = 6x + 4y - x^2 - y^2 + \lambda(4 - x - y).$$

First-order conditions

$$6 - 2x - \lambda = 0, \quad 4 - 2y - \lambda = 0, \quad x + y = 4.$$

These are linear in (x, y, λ) :

$$\begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 1 \\ 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ \lambda \end{bmatrix} = \begin{bmatrix} 6 \\ 4 \\ 4 \end{bmatrix}.$$

Solve

Eliminate as in Linear Algebra: subtracting the second equation from the first removes λ :

$2 - 2x + 2y = 0$, so $x - y = 1$. With $x + y = 4$,

$$x^* = \frac{5}{2}, \quad y^* = \frac{3}{2}, \quad \lambda^* = 6 - 2x^* = 1.$$

Why this is the maximum

$H_f = \text{diag}(-2, -2) \prec 0$: f is strictly concave and the constraint is affine, so the FOCs are sufficient and (x^*, y^*) is the unique global maximizer, with value $\frac{25}{2}$.

Four steps in every Lagrange problem: write $\mathcal{L}$, differentiate, solve with the constraint, justify optimality. Substituting an easy constraint into f is quicker.

The constrained problem as a curve on a surface

The worked problem again:

$$\max_{x,y} 6x + 4y - x^2 - y^2 \quad \text{s.t.} \quad x + y = 4.$$

- ◆ The surface is $z = f(x, y)$, a dome.
- ◆ The constraint is a line in the x - y plane (green). Lifting it to the surface gives the red curve $\{(x, 4 - x, f(x, 4 - x))\}$.
- ◆ Along that curve $f(x, 4 - x) = 10x - 2x^2$, whose highest point is $(\frac{5}{2}, \frac{3}{2})$ with value $\frac{25}{2}$.

Constrained maximization is the search for the highest point of the lifted curve.

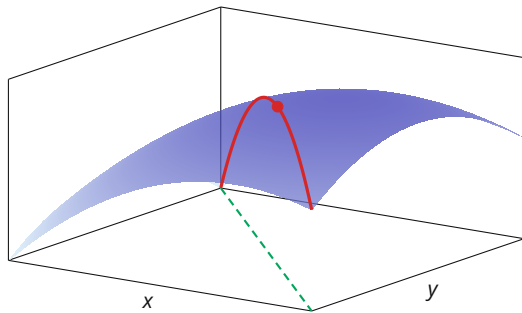


Figure 12 · The objective surface, the constraint line in the base plane, the constraint lifted onto the surface, and its highest point. The next page projects this picture back to the plane, where the highest attainable level curve is tangent to the constraint.

Tangency of level curve and constraint

At an interior constrained optimum with $\nabla g(x^*) \neq 0$ and $\nabla f(x^*) \neq 0$, the objective level curve and the constraint curve are tangent:

$$\nabla f(x^*) = \lambda^* \nabla g(x^*).$$

- ◆ The constraint is the level set $g^{-1}(\{b\})$ from Functions, and from Differentiation a gradient is normal to its level set. Tangent level sets have collinear normals.
- ◆ This pins down λ^* , interpreted later as the marginal value of relaxing the constraint.

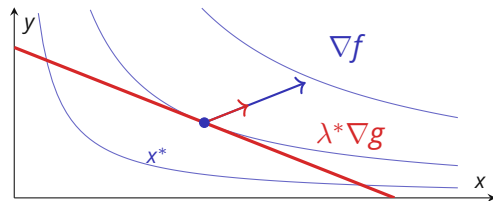


Figure 13 · Level curves of f (blue), the constraint $g = b$ (red), and the collinear gradients at x^* .

The level curve through x^* and the constraint are tangent there.

Second-order conditions along feasible directions

Curvature on the tangent space

For $\max_x f(x)$ s.t. $h(x) = 0$, with $f : \mathbb{R}^n \rightarrow \mathbb{R}$ and $h : \mathbb{R}^n \rightarrow \mathbb{R}^m$ twice continuously differentiable near x^* , let (x^*, λ^*) satisfy the FOCs and suppose $Dh(x^*)$ has full row rank. Write $H_{\mathcal{L}} = \nabla_{xx}^2 \mathcal{L}(x^*, \lambda^*)$. The first-order feasible directions form the tangent space

$$T(x^*) = \ker Dh(x^*) = \{d : Dh(x^*)d = 0\}.$$

Necessary for max $d^\top H_{\mathcal{L}} d \leq 0$ for every $d \in T(x^*)$.

Sufficient for max $d^\top H_{\mathcal{L}} d < 0$ for every nonzero $d \in T(x^*)$.

Unconstrained SOC test curvature in **every direction**; equality-constrained SOC test only the **feasible tangent directions** $d \in T(x^*)$. The bordered Hessian is a computational device for this test and is left to the formal Optimization course.

Further reading: Boyd, §19.2; MIT 6.7220, Lecture 7.

Linear Algebra applied: feasible directions

Kernel and normal space

For affine constraints $Ax = b$, a displacement d remains feasible precisely when

$$A(x + d) = b \quad \Leftrightarrow \quad Ad = 0.$$

Hence feasible directions form $\ker(A)$. With $\mathcal{L}(x, \lambda) = f(x) + \lambda^\top (b - Ax)$, stationarity gives

$$\nabla f(x^*) = A^\top \lambda^*.$$

Therefore, for every feasible direction d ,

$$\nabla f(x^*)^\top d = (\lambda^*)^\top Ad = 0.$$

Thus $\nabla f(x^*)$ is orthogonal to every feasible direction $d \in \ker(A)$.

Feasible directions and restricted curvature

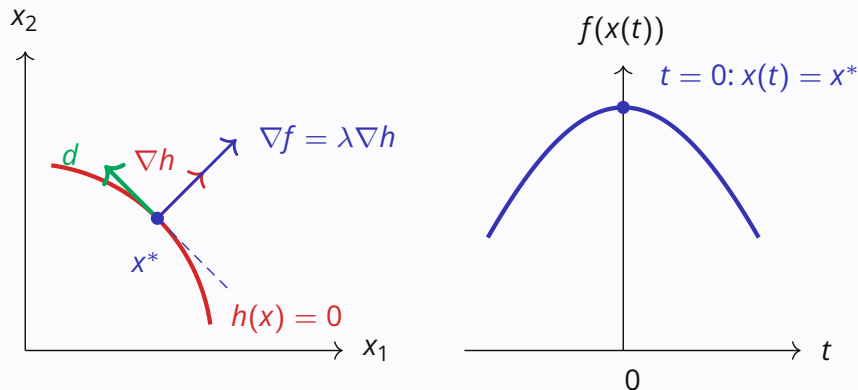


Figure 14 · Left: at x^* the tangent direction d satisfies $Dh(x^*)d = 0$, ∇h and ∇f are collinear, and the dashed level curve of f is tangent to the constraint. Right: along the constraint, $t \mapsto f(x(t))$ peaks at $t = 0$; $d^\top H_{\mathcal{L}} d < 0$ is its curvature there.

An indefinite Hessian with a constrained maximum

Take $f(x, y) = xy$ with $x + y = 10$, the product problem of the exercises.

- ◆ $H_f = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ is indefinite: along $y = x$ the surface rises as $z = x^2$ (blue curve).
- ◆ The feasible directions satisfy $d_1 + d_2 = 0$. Along $x + y = 10$ the surface falls away from the centre as $z = x(10 - x)$ (red curve), with its highest point $(5, 5, 25)$.
- ◆ The constrained test asks for the sign of $d^\top H_{\mathcal{L}} d$ on the tangent line alone, where it is $-2t^2 < 0$.

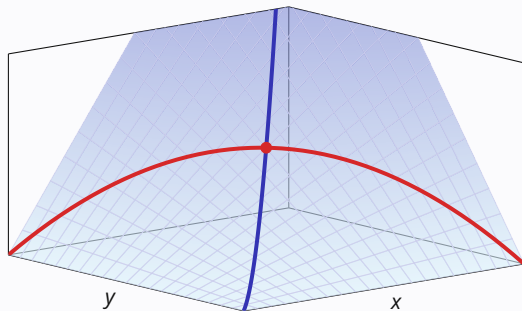


Figure 15 · The saddle $z = xy$, the rising section along $y = x$, the falling section along the constraint $x + y = 10$, and the constrained maximizer $(5, 5)$. One surface has a rising and a falling direction; the constraint selects the direction that decides the problem.

Quadratic Lagrange equations

A block linear system

Consider

$$\min_x \frac{1}{2}x^\top Qx - c^\top x \quad \text{s.t.} \quad Ax = b,$$

with $Q = Q^\top$, and use $\mathcal{L} = \frac{1}{2}x^\top Qx - c^\top x + \lambda^\top (Ax - b)$. The stationarity and feasibility equations are

$$Qx - c + A^\top \lambda = 0, \quad Ax = b.$$

They form the linear system

$$\begin{bmatrix} Q & A^\top \\ A & 0 \end{bmatrix} \begin{bmatrix} x \\ \lambda \end{bmatrix} = \begin{bmatrix} c \\ b \end{bmatrix}.$$

Additional example: output on a technology frontier

A firm chooses positive inputs (x, y) and production is $Q(x, y) = x^2 y$. The feasible frontier is

$$2x^2 + y^2 = 3.$$

The Lagrangian is

$$\mathcal{L} = x^2 y + \lambda(3 - 2x^2 - y^2),$$

with first-order conditions

$$2xy - 4\lambda x = 0, \quad x^2 - 2\lambda y = 0, \quad 2x^2 + y^2 = 3.$$

With $x > 0$ the first equation gives $\lambda = y/2$, the second then gives $x^2 = y^2$, so $x = y$, and the frontier gives $x^* = y^* = 1$, $\lambda^* = \frac{1}{2}$. Zero input gives zero output on the boundary, so $(1, 1)$ is the maximizer.

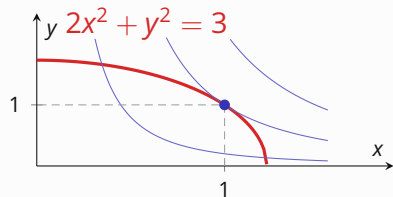


Figure 16 · Isoquants $x^2 y = c$ (blue) and the frontier (red), tangent at $(1, 1)$.

Restricted least squares

Constrained least-squares

The normal equations $X^\top X \hat{\beta} = X^\top y$ of the Second-order conditions section solve the unrestricted problem. Econometrics often imposes linear equality restrictions on coefficients:

$$y = X\beta + u, \quad R\beta = r.$$

The constrained least-squares problem is

$$\min_{\beta} (y - X\beta)'(y - X\beta) \quad \text{s.t.} \quad R\beta = r.$$

Examples include equal treatment effects, zero long-run restrictions, and adding-up restrictions. For instance, with two coefficients, $\beta_1 = \beta_2$ is $R = \begin{bmatrix} 1 & -1 \end{bmatrix}$ and $r = 0$.

Likelihood ratio test

Restricted likelihood

A likelihood-ratio test compares the best unrestricted fit with the best fit under restrictions:

$$LR = 2(\ell(\hat{\theta}) - \ell(\hat{\theta}_R)).$$

Unrestricted. Maximize $\ell(\theta)$ over all admissible θ .

Restricted. Maximize $\ell(\theta)$ subject to $R\theta = r$.

The statistic measures how much optimized value is lost when the equality restrictions are imposed. For instance, $\ell(\hat{\theta}) = -100$ and $\ell(\hat{\theta}_R) = -103$ give $LR = 2[-100 - (-103)] = 6$.

Lagrange multipliers

Solve both problems with a Lagrangian: write $\mathcal{L}$, solve the first-order conditions together with the constraint, report (x^*, y^*, λ^*) , and say why the stationary point is the global optimum.

(a) $\max_{x,y>0} xy$ s.t. $x + y = 10$.

(b) $\min_{x,y \in \mathbb{R}} x^2 + y^2$ s.t. $x + 2y = 4$.

The product problem

For (a), use $\mathcal{L} = xy + \lambda(10 - x - y)$. The FOCs are $y - \lambda = 0$ and $x - \lambda = 0$, so $x = y = \lambda$; the constraint then gives

$$(x^*, y^*, \lambda^*) = (5, 5, 5).$$

The Hessian of xy is $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, indefinite, so concavity is unavailable as the reason for optimality. Substitute the constraint instead: on the feasible set $y = 10 - x$ with $0 < x < 10$, the objective is $\varphi(x) = x(10 - x)$, with $\varphi'' = -2 < 0$, so $x^* = 5$ is its unique maximizer and $(5, 5)$ is the global constrained maximizer.

Distance to a line

For (b), use $\mathcal{L} = x^2 + y^2 + \lambda(4 - x - 2y)$. The FOCs are $2x - \lambda = 0$ and $2y - 2\lambda = 0$, so $x = \lambda/2$ and $y = \lambda$; the constraint gives $\lambda/2 + 2\lambda = 4$. Hence

$$(x^*, y^*, \lambda^*) = \left(\frac{4}{5}, \frac{8}{5}, \frac{8}{5} \right).$$

The objective is strictly convex and the constraint is affine, so by the proposition on the Lagrangian page this feasible solution of the FOCs is the unique global minimizer. The vector $(x^*, y^*) = \frac{4}{5}(1, 2)$ is parallel to the constraint normal: the point of the line nearest the origin lies on the normal through the origin.

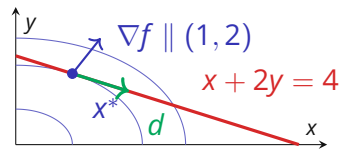


Figure 17 · Level circles of $x^2 + y^2$, the constraint line, the normal direction $(1, 2)$, and the direction $(2, -1)$ along the line.

Nash bargaining

The product problem in game theory

In a two-player bargaining problem, let $d = (d_1, d_2)$ be the disagreement point. The Nash bargaining solution maximizes the product of the players' gains from agreement, $\max_{u \in S, u \geq d} (u_1 - d_1)(u_2 - d_2)$. If $d = (0, 0)$ and the Pareto frontier is $u_1 + u_2 = 10$ with $u_1, u_2 > 0$, this becomes

$$\max_{u_1, u_2 > 0} u_1 u_2 \quad \text{s.t.} \quad u_1 + u_2 = 10,$$

the product problem solved above. Hence $(u_1^*, u_2^*) = (5, 5)$.

Geometry. The Nash product is the area of the rectangle between d and u . Here the sum of its side lengths is fixed, so the maximum-area rectangle is the square 5×5 .

The equal split comes from the symmetry of this particular feasible frontier and disagreement point. Equal utility levels are specific to $d = (0, 0)$; in this linear-surplus example the Nash solution equalizes the gains $u_i - d_i$ from disagreement, so with $d = (2, 0)$ it is $(6, 4)$.

Further reading: Nash (1950), *The Bargaining Problem*, *Econometrica* 18, pp. 155–162.

Lagrange multipliers with a nonlinear constraint

Both problems of the preceding exercise have an affine constraint, so ∇g is the same vector at every feasible point. Here the feasible set is a circle and ∇g turns along it.

- (a) Solve $\max_{x,y>0} xy$ s.t. $x^2 + y^2 = 2$: write $\mathcal{L}$, solve the first-order conditions together with the constraint, and report (x^*, y^*, λ^*) .
- (b) Substitute $x = \sqrt{2} \cos t, y = \sqrt{2} \sin t$ with $t \in (0, \pi/2)$, maximize the resulting function of one variable, and compare with (a).
- (c) Replace the right side by $c > 0$. Find the value $V(c)$ and check $V'(c) = \lambda^*(c)$.

Lagrange multipliers with a nonlinear constraint

With $\mathcal{L} = xy + \lambda(2 - x^2 - y^2)$, the first-order conditions and the constraint are

$$y - 2\lambda x = 0, \quad x - 2\lambda y = 0, \quad x^2 + y^2 = 2.$$

(a) Multiplying the first by y and the second by x gives $y^2 = 2\lambda xy = x^2$, so $x = y$ for $x, y > 0$, and $2x^2 = 2$ gives

$$(x^*, y^*) = (1, 1), \quad \lambda^* = \frac{1}{2}, \quad x^*y^* = 1.$$

(b) On the circle, $xy = 2 \cos t \sin t = \sin 2t$, which rises on $(0, \pi/4)$ and falls on $(\pi/4, \pi/2)$. The maximum is at $t = \pi/4$, giving $(1, 1)$ again. Existence follows as well: $\sin 2t$ is continuous on $[0, \pi/2]$ and vanishes at both endpoints, so its maximum there is interior.

(c) The same steps with $x^2 + y^2 = c$ give $x^* = y^* = \sqrt{c/2}$, $\lambda^* = 1/2$ and $V(c) = c/2$, so $V'(c) = 1/2 = \lambda^*$: the multiplier is the rate at which the optimized value responds to the constraint level.

Two equality constraints

One multiplier for each constraint. For a parameter $a \in \mathbb{R}$, consider

$$\min_{x,y,z} x^2 + y^2 + z^2 \quad \text{s.t.} \quad x + y + z = 3, \quad x - y = a,$$

with $\mathcal{L} = x^2 + y^2 + z^2 + \lambda(3 - x - y - z) + \mu(a - x + y)$.

- Solve the three stationarity equations together with both constraints for $(x^*, y^*, z^*, \lambda^*, \mu^*)$ as functions of a .
- Report the solution at $a = 0$ and at $a = 1$, and say what the value of μ^* at $a = 0$ says about the second constraint there.
- Verify $\nabla f(x^*) = \lambda^* \nabla g_1(x^*) + \mu^* \nabla g_2(x^*)$ at $a = 1$, with $g_1(x, y, z) = x + y + z$ and $g_2(x, y, z) = x - y$.
- Find $V(a)$ and check $V'(a) = \mu^*(a)$.

Two equality constraints

Each constraint has its own multiplier. (a) Stationarity in x, y, z gives

$$2x - \lambda - \mu = 0, \quad 2y - \lambda + \mu = 0, \quad 2z - \lambda = 0.$$

Subtracting gives $\mu = x - y = a$, adding gives $\lambda = x + y$, and $z = \lambda/2$. The first constraint becomes $\frac{3}{2}(x + y) = 3$, so $x + y = 2$, $\lambda^* = 2$, $z^* = 1$, and

$$x^* = 1 + \frac{a}{2}, \quad y^* = 1 - \frac{a}{2}, \quad \mu^* = a.$$

Strict convexity with affine constraints makes this the unique global minimizer for every a . **(b)**

At $a = 0$: $(1, 1, 1)$, $\lambda^* = 2$, $\mu^* = 0$; the first constraint alone selects a point with $x = y$, so the second has no effect on the value. At $a = 1$: $(\frac{3}{2}, \frac{1}{2}, 1)$, $\lambda^* = 2$, $\mu^* = 1$.

(c) At $a = 1$: $\nabla f = (3, 1, 2)$ and $2(1, 1, 1) + 1 \cdot (1, -1, 0) = (3, 1, 2)$. **(d)** $V(a) = 3 + \frac{a^2}{2}$, so $V'(a) = a = \mu^*(a)$.

Feasible directions and the block system

Second-order checks for the two Lagrange problems

For the two problems of the preceding exercise:

- (a) Find a nonzero feasible direction d with $Dh(x^*)d = 0$ and evaluate $d^\top H_{\mathcal{L}}(x^*, \lambda^*)d$.
What does the sign confirm in each case?
- (b) Write the FOCs of the distance problem as a block linear system in (x, y, λ) .

Feasible directions and the block system

Restricted curvature has the right sign in both problems

Product problem. $H_{\mathcal{L}} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ and feasible directions satisfy $d_1 + d_2 = 0$. With $d = (t, -t)$,

$$d^{\top} H_{\mathcal{L}} d = -2t^2 < 0 \quad (t \neq 0),$$

confirming a strict local maximum (global by the solution page). Here $H_{\mathcal{L}} = H_f$, since the constraint is affine, and H_f is indefinite on $\mathbb{R}^2$: the unconstrained test classifies $(5, 5)$ as a saddle of f . The constrained test asks for the sign of the same quadratic form on the tangent line $d_1 + d_2 = 0$ alone, where it is negative.

Distance problem. Feasible directions form $\ker[1 \ 2] = \text{span}\{(2, -1)\}$, the direction drawn in Figure 17. With $d = (2t, -t)$ and $H_{\mathcal{L}} = 2I$, $d^{\top} H_{\mathcal{L}} d = 10t^2 > 0$: a strict local minimum (global by the Lagrangian page). Stationarity and the constraint form

$$\begin{bmatrix} 2 & 0 & -1 \\ 0 & 2 & -2 \\ 1 & 2 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ \lambda \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 4 \end{bmatrix}.$$

This is the block system of the quadratic Lagrange page with $Q = 2I$, $A = [1 \ 2]$, $c = 0$, and $b = 4$. The multiplier column has the opposite sign because our Lagrangian uses $\lambda(b - Ax)$.

Section 4

Inequality constraints (KKT)

Binding or slack

Problem. $\max_{x \in \mathbb{R}^n} f(x)$ subject to $g_j(x) \leq b_j$.

At an optimum each constraint is either

Binding $g_j(x) = b_j$, or

Slack $g_j(x) < b_j$.

Resource constraint $x + y \leq 10$: plan (4, 6) binds; plan (3, 5) uses eight of the ten units and is slack.

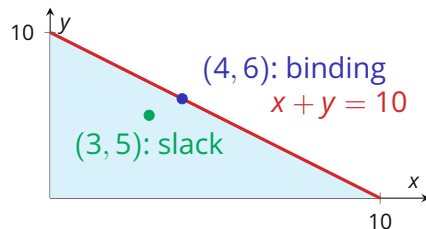


Figure 18 · The constraint $x + y \leq 10$: (4, 6) lies on the boundary, (3, 5) inside.

When an inequality must bind

Proposition. Consider $\max_{x \geq 0} f(x)$ subject to $g(x) \leq b$, with g continuous. If f is strictly increasing in at least one coordinate, then every maximizer satisfies $g(x^*) = b$: the problems with $g(x) \leq b$ and with $g(x) = b$ have the **same maximizers**.

Affine case. For $g(x) = a^\top x$ with $a \gg 0$ the same ε can be written down, $\varepsilon = (b - a^\top x^*)/(2a_i)$, giving $a^\top (x^* + \varepsilon e_i) = (a^\top x^* + b)/2 < b$; this is the step behind the budget binding on the consumer pages. For $g(x) = x_1^2 + x_2^2$ it applies unchanged.

Other inequalities may stay slack. Another active constraint can block every improving move: in $\max_{x \geq 0} x$ s.t. $x^2 \leq 100$, $x \leq 1$ the maximizer is $x^* = 1$ and the first constraint holds strictly.

Further reading: UC Berkeley Econ 101A, Lecture 5, p. 17; Levin, *Consumer Theory*, Proposition 4, pp. 4–5.

Microeconomics applied: constraints that must bind

Budget exhaustion, no overproduction, no excess utility

Utility maximization. $\max_{x \geq 0} u(x)$ s.t.
 $p \cdot x \leq w, p \gg 0$. With u strictly increasing in one good, or locally non-satiated preferences, every maximizer spends the whole budget: $p \cdot x^* = w$ (Walras' Law).

The same argument gives no excess utility for expenditure minimization when $\bar{u}$ exceeds the utility of the zero bundle. The envelope pages use all three equalities:

$$\text{Marshallian: } p \cdot x \leq w \quad \Rightarrow \quad p \cdot x^* = w,$$

$$\text{cost minimization: } F(x) \geq q \quad \Rightarrow \quad F(x^*) = q,$$

$$\text{Hicksian: } u(x) \geq \bar{u} \quad \Rightarrow \quad u(x^*) = \bar{u}.$$

Cost minimization. $\min_{L, K \geq 0} wL + rK$ s.t.
 $F(L, K) \geq q$, with $w, r > 0$, F continuous and $F(0, 0) < q$. A minimizer never overproduces, $F(L^*, K^*) = q$: with $F(L^*, K^*) > q$ some input is positive, and continuity lets a small cut in it keep $F \geq q$ while the cost falls.

Further reading: [Levin, Consumer Theory](#), Prop. 4, pp. 4–5; MIT 14.121, [Consumer Theory slides](#), pp. 11 and 29; [DellaVigna, Econ 101A Lecture 16](#), p. 6.

KKT conditions

For $\max_x f(x)$ subject to $g_j(x) \leq b_j$, use $\mathcal{L}(x, \lambda) = f(x) + \sum_j \lambda_j(b_j - g_j(x))$. **In this maximization convention the multiplier multiplies $b_j - g_j$ and $\lambda_j \geq 0$.** The multiplier of an equality constraint is **free in sign**; an inequality bounds one direction. At a local optimum where a constraint qualification holds, there are multipliers $\lambda^* \geq 0$ satisfying

Four conditions.

Stationarity	$\nabla_x \mathcal{L}(x^*, \lambda^*) = 0,$
Feasibility	$g_j(x^*) \leq b_j,$
Dual feasibility	$\lambda_j^* \geq 0,$
Complementary slackness	$\lambda_j^*(b_j - g_j(x^*)) = 0.$

Video: [Arizona Math Camp, Lecture 40\(B\): The Kuhn–Tucker Conditions and Theorem.](#)

Normals of binding linear constraints

The geometry of stationarity

Suppose the constraints are $a_j^\top x \leq b_j$, so that $g_j(x) = a_j^\top x$ has $\nabla g_j(x) = a_j$ at every point. KKT stationarity becomes

$$\nabla f(x^*) = \sum_{j=1}^m \lambda_j^* \nabla g_j(x^*) = \sum_{j=1}^m \lambda_j^* a_j, \quad \lambda_j^* \geq 0.$$

Complementary slackness sets $\lambda_j^* = 0$ whenever constraint j is slack. Thus only normals of binding constraints can contribute to the objective gradient. At a corner, several normals can be needed; at a smooth boundary point, one normal may suffice.

If a feasible direction d points into every binding constraint, then $a_j^\top d \leq 0$. Hence $\nabla f(x^*)^\top d \leq 0$: moving a small distance along d cannot raise the objective to first order.

Objective gradient in the cone of binding normals

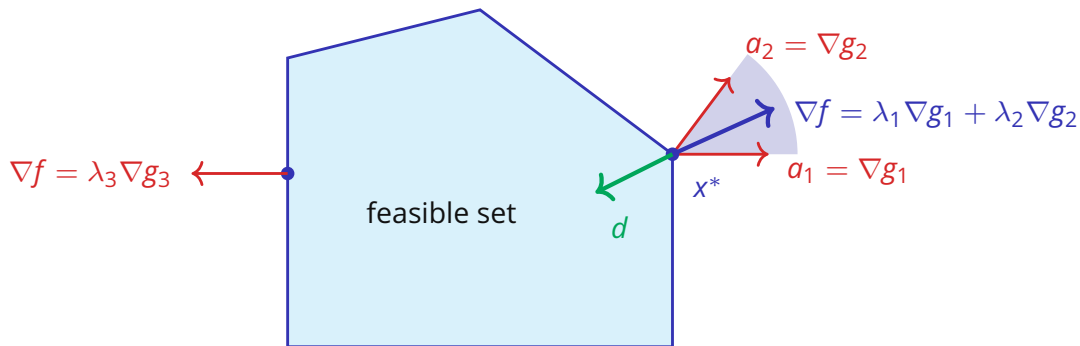


Figure 19 · At the corner x^* two constraints bind and ∇f lies in the shaded cone spanned by their gradients $\nabla g_1, \nabla g_2$; at a smooth boundary point one normal suffices. A feasible direction d points into the set and makes an obtuse angle with ∇f .

Video: Arizona Math Camp 40(A), geometric insight.

Complementary slackness

A product is zero if and only if one of its factors is zero, so $\lambda_j^* (b_j - g_j(x^*)) = 0$ holds if and only if $\lambda_j^* = 0 \vee g_j(x^*) = b_j$, with the inclusive “or” from Preliminaries: both can hold. Equivalently, $g_j(x^*) < b_j \Rightarrow \lambda_j^* = 0$ and $\lambda_j^* > 0 \Rightarrow g_j(x^*) = b_j$. **The converses fail:** a binding constraint may have a zero multiplier (Figure 20, right panel).

Duality theory: the saddle-point view

Suppose strong duality holds and both optima are attained. The primal and dual solutions are then a saddle point of the Lagrangian: the primal variables maximize it, the multipliers minimize it. The primal and dual problems, the dual function and strong duality are developed in the formal Optimization course.

Further reading: Boyd, Ch. 18; Boyd–Vandenberghe, §5.4.

Three ways complementary slackness can look

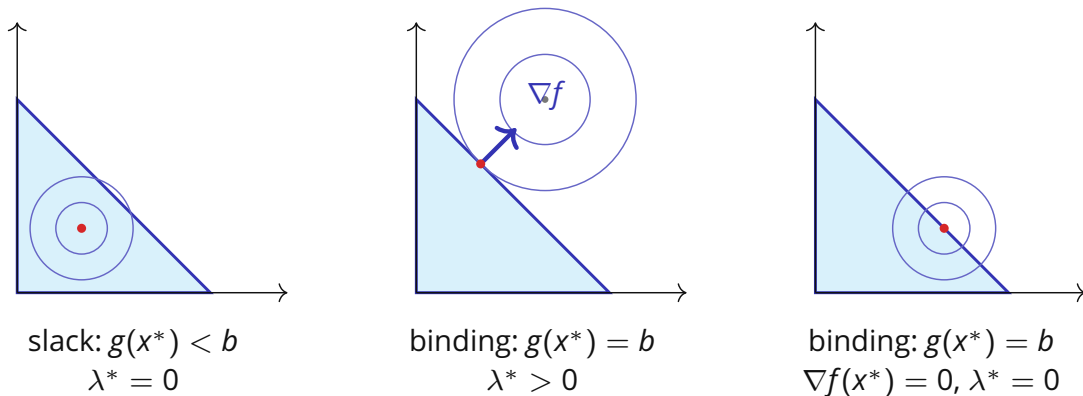


Figure 20 · Level circles of f over the same feasible triangle. Left: the peak is interior, the constraint is slack, the multiplier is zero. Middle: the peak lies outside, the optimum is pushed onto the constraint and the multiplier is positive. Right: the peak lies on the constraint line itself, so the constraint binds with a zero multiplier.

Nonnegativity: two equivalent KKT forms

Suppose the problem also requires $x_i \geq 0$. Write this as $-x_i \leq 0$ and let $\tilde{\mathcal{L}}(x, \lambda) = f(x) + \sum_j \lambda_j (b_j - g_j(x))$, omitting only the nonnegativity constraint.

With the multiplier μ_i .

$$\begin{aligned}\mathcal{L} &= \tilde{\mathcal{L}} + \mu_i x_i, \\ \frac{\partial \tilde{\mathcal{L}}}{\partial x_i} + \mu_i &= 0, \\ x_i \geq 0, \quad \mu_i &\geq 0, \quad \mu_i x_i = 0.\end{aligned}$$

Stationarity is an equality.

After eliminating μ_i . Since $\mu_i = -\partial \tilde{\mathcal{L}} / \partial x_i$,

$$\begin{aligned}x_i \geq 0, \quad \frac{\partial \tilde{\mathcal{L}}}{\partial x_i} &\leq 0, \\ x_i \frac{\partial \tilde{\mathcal{L}}}{\partial x_i} &= 0.\end{aligned}$$

If $x_i > 0$, the partial derivative is **zero**. If $x_i = 0$, it may be **negative**.

Further reading: Columbia Math Camp, *Constrained Optimization*, Sec. 6.

Constraint qualifications for KKT

Beyond the equality case

The equality pages already used a constraint qualification, $\text{rank } Dg(x^*) = m$. Once inequalities are present, that role passes to a family of conditions. **KKT necessity is invoked only when one of them holds.**

Each of them rules out irregular local descriptions of the feasible boundary.

Further reading: Boyd, §19.4; MIT 6.7220, Lec. 7; Border, *Summary Notes on Maximization*, §6, pp. 11–12.

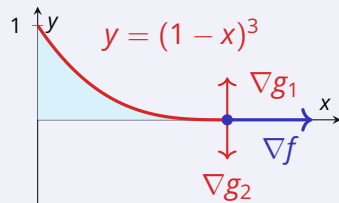


Figure 21 $\cdot \max x \text{ s.t. } y \leq (1-x)^3, y \geq 0$: the optimum is the cusp $(1,0)$; $\nabla g_1 = (0,1)$ and $\nabla g_2 = (0,-1)$ are vertical, so their cone misses $\nabla f = (1,0)$.

Video: Stanford EE364A, Lecture 8: Lagrangian duality and Slater's condition.

LICQ, MFCQ and Slater

Three constraint qualifications

For $\max_x f(x)$ subject to $g_j(x) \leq b_j$ and $h_l(x) = c_l$, $l = 1, \dots, m$, write $A(x^*) = \{j : g_j(x^*) = b_j\}$ for the inequalities that bind at x^* .

LICQ the gradients $\nabla h_l(x^*)$ and $\nabla g_j(x^*)$, $j \in A(x^*)$, are linearly independent.

MFCQ the $\nabla h_l(x^*)$ are linearly independent and some d has $\nabla h_l(x^*)^\top d = 0$ for every l and $\nabla g_j(x^*)^\top d < 0$ for every $j \in A(x^*)$.

Slater for concave f , convex g_j and affine h_l , some feasible $\bar{x}$ has $g_j(\bar{x}) < b_j$ for every j .

With no inequality present LICQ is the rank condition of the equality pages, and LICQ implies MFCQ. Slater constrains the problem; the other two are checked at x^* . For $\max x$ s.t. $x^2 = 0$, $g'(0) = 0$ fails both gradient conditions.

Further reading: Burke, Def. 5; MIT 6.7220, Lec. 7; Boyd and Vandenberghe, §5.2.3.

A regular description of the same optimum

The cone of binding normals when the boundary is smooth

Keep the objective $\max x$ and the optimum $(1, 0)$ of Figure 21, and replace the cusp by a linear boundary:

$$\max_{x,y} x \quad \text{s.t.} \quad x + y \leq 1, \quad y \geq 0.$$

Feasibility gives $x \leq 1$, so $(1, 0)$ is again the optimum. With $\mathcal{L} = x + \lambda_1(1 - x - y) + \lambda_2 y$, stationarity gives

$$1 - \lambda_1 = 0, \quad -\lambda_1 + \lambda_2 = 0,$$

so $\lambda_1^* = \lambda_2^* = 1$, both nonnegative: the KKT system has a solution here.

The binding normals $\nabla g_1 = (1, 1)$ and $\nabla g_2 = (0, -1)$ span a cone containing $\nabla f = (1, 0) = (1, 1) + (0, -1)$. In Figure 21 the same objective has the same optimum, and there the binding normals $(0, 1)$ and $(0, -1)$ span vertical vectors alone. The two problems differ in how the boundary is written near $(1, 0)$, which a constraint qualification restricts.

Second-order conditions at a KKT point

Curvature on the critical cone

For $\max_x f(x)$ s.t. $g_j(x) \leq b_j$, let (x^*, λ^*) satisfy KKT with $\mathcal{L} = f + \sum_j \lambda_j(b_j - g_j)$, and write $H_{\mathcal{L}} = \nabla_{xx}^2 \mathcal{L}(x^*, \lambda^*)$. Split the active set $A(x^*) = \{j : g_j(x^*) = b_j\}$ by the multiplier:

$$A_+ = \{j \in A(x^*) : \lambda_j^* > 0\}, \quad A_0 = \{j \in A(x^*) : \lambda_j^* = 0\}.$$

The **critical cone** holds the first-order feasible directions along which the objective is flat to first order:

$$C(x^*, \lambda^*) = \{d : \nabla g_j(x^*)^\top d = 0 \ (j \in A_+), \quad \nabla g_j(x^*)^\top d \leq 0 \ (j \in A_0)\}.$$

Sufficient for a strict local max $d^\top H_{\mathcal{L}} d < 0$ for every nonzero $d \in C(x^*, \lambda^*)$.

Necessary at a local max with a constraint qualification, $d^\top H_{\mathcal{L}} d \leq 0$ on the directions tangent to the active constraints.

When KKT conditions are sufficient

Two routes lead from a KKT point to optimality. The local route tests curvature on the feasible directions at x^* ; the global route, below, asks for concavity of the objective with convex constraint functions, under which a KKT point is a global maximum.

If (x^*, λ) satisfies KKT, **f is concave and each g_j is convex**, and the sign convention is $\mathcal{L} = f + \sum_j \lambda_j(b_j - g_j)$, then x^* is a **global** maximizer. For $\min f$ with **convex f** and **convex g_j** , and $\mathcal{L} = f - \sum_j \lambda_j(b_j - g_j)$, the KKT conditions give a **global** minimizer. Strict concavity, or strict convexity, makes the optimizer unique.

Convexity applied. For convex g_j , each constraint set is the convex lower contour set $L_{g_j}(b_j) = \{x : g_j(x) \leq b_j\}$. Intersections of convex sets are convex, so the feasible set is convex. Concavity of f then turns the first-order inequalities into a global conclusion, as the next page shows.

Why a KKT point is then a global maximum

Let y be feasible. Concavity of f puts its graph below the tangent plane at x^* ,

$$f(y) \leq f(x^*) + \nabla f(x^*)^\top (y - x^*),$$

and KKT stationarity replaces the gradient by $\nabla f(x^*) = \sum_j \lambda_j^* \nabla g_j(x^*)$. Convexity of each g_j puts its graph above its own tangent plane,

$$\nabla g_j(x^*)^\top (y - x^*) \leq g_j(y) - g_j(x^*),$$

and $\lambda_j^* \geq 0$ preserves each inequality, so

$$f(y) - f(x^*) \leq \sum_j \lambda_j^* [g_j(y) - g_j(x^*)] \leq \sum_j \lambda_j^* [b_j - g_j(x^*)] = 0,$$

where the middle step uses $g_j(y) \leq b_j$ and the last uses complementary slackness term by term.

Hence $f(y) \leq f(x^*)$ at every feasible y , which is what a global maximum means. Concavity gives the first inequality, convexity and $\lambda_j^* \geq 0$ the second, feasibility of y the third, complementary slackness the last equality.

KKT sufficiency beyond concavity

Quasiconcave programming

For the convention $g_j(x) \leq b_j$, suppose f and the g_j are differentiable and each g_j is quasiconvex. If (x^*, λ^*) satisfies KKT, then x^* is a global maximizer when either

- (a) f is concave; or
- (b) f is quasiconcave and $\nabla f(x^*) \neq 0$.

The requirement $\nabla f(x^*) \neq 0$ in (b) is imposed at the KKT point. It excludes stationary points at which quasiconcavity gives no first-order ranking of feasible alternatives; the next page shows one.

Further reading: Walker, *Nonlinear Programming and the Kuhn–Tucker Conditions*.

Pseudoconcavity and first-order sufficiency

Pseudoconcavity

On a convex domain S , a differentiable $f : S \rightarrow \mathbb{R}$ is pseudoconcave when, for every $x, y \in S$,

$$f(y) > f(x) \Rightarrow \nabla f(x)^\top (y - x) > 0.$$

This implication directly converts the KKT gradient inequality for a feasible y into $f(y) \leq f(x^*)$. Hence a pseudoconcave objective and quasiconvex constraint functions also make KKT sufficient. Walker's KKT theorem needs $\nabla f(x^*) \neq 0$ only at the KKT point; his characterization of pseudoconcavity from quasiconcavity assumes $\nabla f(x) \neq 0$ throughout the domain.

Two comparisons. $f(x) = e^x$ is strictly convex, so it fails concavity, yet it is pseudoconcave on $\mathbb{R}$: $f(y) > f(x)$ forces $y > x$, and then $f'(x)(y - x) = e^x(y - x) > 0$. $f(x) = x^3$ is quasiconcave but fails pseudoconcavity at 0: $f(1) > f(0)$ while $f'(0)(1 - 0) = 0$. Its stationary point 0 satisfies KKT for $\max x^3$ and is beaten by every $x > 0$: the case that (b) on the previous page rules out.

Further reading: Walker, *Differentiable Quasiconcave Functions*.

Minimization: the same system in the other sign convention

Standard minimization form and its KKT system

$$\begin{array}{ll} \min_x & f(x) \\ \text{s.t.} & g_i(x) \leq 0, \quad h_r(x) = 0. \end{array} \quad \mathcal{L} = f(x) + \sum_i \lambda_i g_i(x) + \sum_r \nu_r h_r(x), \quad \lambda_i \geq 0.$$

KKT system in the $g_i \leq 0$ form.

$$\begin{aligned} \nabla f(x^*) + \sum_i \lambda_i^* \nabla g_i(x^*) + \sum_r \nu_r^* \nabla h_r(x^*) &= 0, \\ g_i(x^*) \leq 0, \quad h_r(x^*) &= 0, \quad \lambda_i^* \geq 0, \quad \nu_r^* \in \mathbb{R}, \quad \lambda_i^* g_i(x^*) = 0. \end{aligned}$$

Minimizing f is maximizing $-f$ on the same feasible set, so this is the earlier system with the gradient and multipliers relabelled.

Geometry under the two conventions

Active inequalities and the cone they span

Write every inequality with a zero right side: the form $g_j(x) \leq b_j$ of the earlier pages becomes $\tilde{g}_j(x) = g_j(x) - b_j \leq 0$, and the gradients are untouched, $\nabla \tilde{g}_j = \nabla g_j$. With equality constraints $h_r(x) = 0$ also present, collect the active inequalities and the directions the multipliers produce,

$$A(x^*) = \{i : g_i(x^*) = 0\}, \quad N(x^*) = \text{cone}\{\nabla g_i(x^*) : i \in A(x^*)\} + \text{span}\{\nabla h_r(x^*) : r = 1, \dots, p\}.$$

problem with $g_i(x) \leq 0, h_r(x) = 0$	geometry at x^*
$\min_x f(x)$	$-\nabla f(x^*) \in N(x^*)$
$\max_x f(x)$	$\nabla f(x^*) \in N(x^*)$

Complementary slackness gives $\lambda_i^* g_i(x^*) = 0$, so only active inequalities reach the cone; equalities always bind and reach the span with ν_r^* **unrestricted in sign**.

Minimization: a solved KKT example

Problem and Lagrangian

$$\begin{aligned} \min_{x,y} \quad & (x-1)^2 + (y-1)^2 \\ \text{s.t.} \quad & x+y-3 \geq 0, \quad x, y \geq 0. \end{aligned}$$

Let $c(x,y) = x + y - 3$. Then

$$\mathcal{L} = (x-1)^2 + (y-1)^2 - \lambda c(x,y) - \mu_x x - \mu_y y.$$

Either $c(x,y) > 0$ or $c(x,y) = 0$. The next page rules out the first case, so the constraint binds at the solution.

Further reading: Tibshirani, CMU 10-725, *Karush-Kuhn-Tucker conditions*.

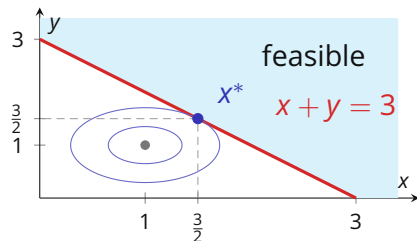


Figure 22 · Level circles around the target $(1, 1)$, the feasible half-plane $x + y \geq 3$ (shaded), and the nearest feasible point x^* .

Minimization: ruling out the slack case

Stationarity for this problem is

$$2(x - 1) - \lambda - \mu_x = 0, \quad 2(y - 1) - \lambda - \mu_y = 0, \quad \lambda c = \mu_x x = \mu_y y = 0.$$

Suppose $c(x, y) > 0$. Complementary slackness gives $\lambda = 0$, so $\mu_x = 2(x - 1)$ and $\mu_y = 2(y - 1)$. A candidate at $x = 0$ would need $\mu_x = -2$, so $x > 0$, hence $\mu_x = 0$ and $x = 1$; the same three steps give $y = 1$. That candidate has $x + y = 2 < 3$, which fails the constraint $x + y \geq 3$.

Therefore $c(x, y) = 0$. In this remaining case a candidate at $x = 0$ would need $\mu_x = -2 - \lambda < 0$, and one at $y = 0$ would need $\mu_y = -2 - \lambda < 0$, so $x, y > 0$ and complementary slackness gives $\mu_x = \mu_y = 0$.

Minimization: KKT calculation

With $\mu_x = \mu_y = 0$, stationarity and the binding constraint give

$$2(x - 1) - \lambda = 0, \quad 2(y - 1) - \lambda = 0, \quad x + y = 3.$$

The first two equations and $x + y = 3$ give

$$x = y = 1 + \frac{\lambda}{2}, \quad 3 = x + y = 2 + \lambda \Rightarrow \lambda = 1.$$

Substitution gives

$$(x^*, y^*) = \left(\frac{3}{2}, \frac{3}{2}\right), \quad \lambda^* = 1, \quad \mu_x^* = \mu_y^* = 0.$$

Primal and dual feasibility and complementary slackness hold. The objective is convex and the constraints define a convex feasible set, so this KKT point is the global minimizer.

Consumer choice: solving for the multipliers

Consumer problem The consumer has perfect-substitutes utility:

$$\begin{aligned} \max_{x,y \geq 0} \quad & \alpha x + \beta y \\ \text{subject to} \quad & p_x x + p_y y \leq m, \end{aligned}$$

where $\alpha, \beta, p_x, p_y, m > 0$.

Slack-form Lagrangian

$$\mathcal{L} = \alpha x + \beta y + \lambda(m - p_x x - p_y y) + \mu_x x + \mu_y y.$$

Here λ is the marginal utility of income; μ_x and μ_y enforce nonnegativity.

KKT conditions

Stationarity

$$\alpha - \lambda p_x + \mu_x = 0, \quad \beta - \lambda p_y + \mu_y = 0.$$

Primal and dual feasibility

$$x, y \geq 0, \quad p_x x + p_y y \leq m, \quad \lambda, \mu_x, \mu_y \geq 0.$$

Complementary slackness

$$\lambda(m - p_x x - p_y y) = 0, \quad \mu_x x = \mu_y y = 0.$$

Consumer choice: corner solutions

Eliminating μ_x, μ_y as on the nonnegativity page gives

$$x \geq 0, \quad \lambda p_x - \alpha \geq 0, \quad x(\lambda p_x - \alpha) = 0,$$

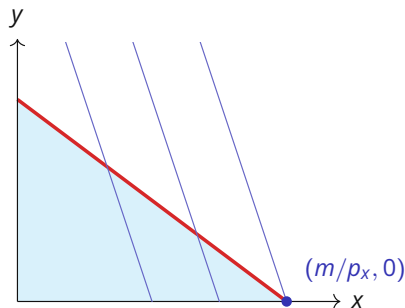
$$y \geq 0, \quad \lambda p_y - \beta \geq 0, \quad y(\lambda p_y - \beta) = 0.$$

Utility is increasing in both goods, so the budget binds, and a consumed good pins down the multiplier:

$$x^* > 0 \Rightarrow \lambda^* = \alpha/p_x, \quad y^* > 0 \Rightarrow \lambda^* = \beta/p_y.$$

At an x-corner, the unused good yields no more utility per dollar:

$$x^* > 0, \quad y^* = 0 \quad \Rightarrow \quad \beta - \lambda^* p_y \leq 0 \quad \Leftrightarrow \quad \frac{\beta}{p_y} \leq \frac{\alpha}{p_x}.$$



$\alpha/p_x > \beta/p_y$: x-corner

Figure 23 · Perfect substitutes: indifference lines $\alpha x + \beta y = k$ (blue) steeper than the budget line (red) put the optimum at the x-corner.

Videos: [Arizona Math Camp 40\(E\), corner solution](#); [Economics in Many Lessons, Utility maximization](#).

Consumer choice: compare the two goods

Compare marginal utility per dollar

$$\begin{aligned} \frac{\alpha}{p_x} &> \frac{\beta}{p_y} &\Rightarrow (x^*, y^*) &= \left(\frac{m}{p_x}, 0 \right), \\ \frac{\alpha}{p_x} &= \frac{\beta}{p_y} &\Rightarrow \{(x, y) \in \mathbb{R}_+^2 : p_x x + p_y y = m\}, \\ \frac{\alpha}{p_x} &< \frac{\beta}{p_y} &\Rightarrow (x^*, y^*) &= \left(0, \frac{m}{p_y} \right). \end{aligned}$$

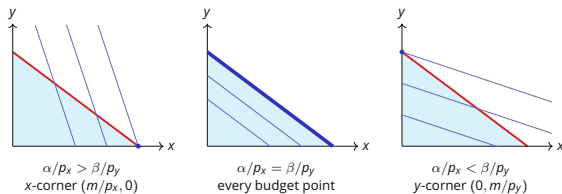


Figure 24 · The tangency condition as slopes: indifference lines $\alpha x + \beta y = k$ (blue) against the budget line (red).

A KKT problem

Choose the feasible point closest to the target $(2, 3)$:

$$\max_{x, y \geq 0} 10 - [(x - 2)^2 + (y - 3)^2]$$

subject to the two resource constraints

$$(1) \ x + y \leq 3, \quad (2) \ x + 2y \leq 6.$$

- ◆ The unconstrained peak $(2, 3)$ is infeasible, so the optimum lies on the boundary of the feasible set.
- ◆ Which of the two resource constraints binds there is what the next two pages settle.

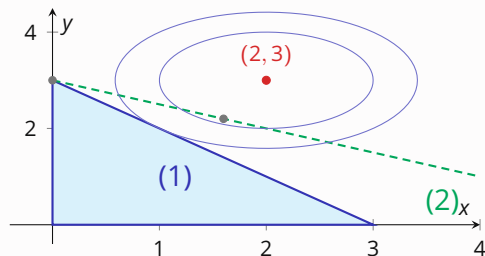


Figure 25 · Level circles of the objective around the target $(2, 3)$, the feasible set, and the two resource constraints. The grey points are the corner and the intersection of the two constraint lines.

The optimum lies on the highest level circle that still meets the feasible set.

Setting up the KKT cases

A case says which inequalities $g_i(x) \leq b_i$ are slack and which bind, and complementary slackness turns that into

$$\text{slack: } g_i(x) < b_i \Rightarrow \lambda_i = 0, \quad \text{binding: } g_i(x) = b_i \Rightarrow \lambda_i \geq 0.$$

A binding constraint still allows $\lambda_i = 0$, so each λ_i is determined by stationarity at the end of a case and never enters the case assumption.

With $\mu_x, \mu_y \geq 0$ on $x \geq 0$ and $y \geq 0$, the conditions for this problem are

$$-2(x - 2) - \lambda_1 - \lambda_2 + \mu_x = 0, \quad -2(y - 3) - \lambda_1 - 2\lambda_2 + \mu_y = 0, \quad \mu_x x = \mu_y y = 0.$$

A candidate with $x > 0$ and $y > 0$ has $\mu_x = \mu_y = 0$ and drops them; one with $x = 0$ or $y = 0$ binds that nonnegativity constraint, and its multiplier stays in stationarity, free to be positive.

Two cases with no candidate

Neither resource constraint binds

The case is $x + y < 3$ and $x + 2y < 6$, so $\lambda_1 = \lambda_2 = 0$ and stationarity gives

$$\mu_x = 2x - 4, \quad \mu_y = 2y - 6.$$

With $x, y > 0$ the candidate is $(x, y) = (2, 3)$, where $x + y = 5 > 3$; at $x = 0$ the first equation forces $\mu_x = -4$ and at $y = 0$ the second forces $\mu_y = -6$. The active set is impossible, so at least one resource constraint binds.

Both resource constraints bind

The case is $x + y = 3$ and $x + 2y = 6$, so $(x, y) = (0, 3)$. Here $y > 0$ gives $\mu_y = 0$, and y -stationarity becomes

$$\lambda_1 + 2\lambda_2 = 0 \quad \Rightarrow \quad \lambda_1 = \lambda_2 = 0,$$

two binding constraints with zero multipliers. The x equation then asks for $4 + \mu_x = 0$, which no $\mu_x \geq 0$ satisfies.

One resource constraint binds

Only the first binds

The case is $x + y = 3$ and $x + 2y < 6$. The slack second constraint gives $\lambda_2 = 0$; the binding first one gives $\lambda_1 \geq 0$ and nothing more yet. With $x, y > 0$, stationarity gives

$$x = 2 - \frac{\lambda_1}{2}, \quad y = 3 - \frac{\lambda_1}{2},$$

and $x + y = 3$ turns into

$$5 - \lambda_1 = 3 \Rightarrow (x, y, \lambda_1) = (1, 2, 2).$$

There $x + 2y = 5 < 6$, so the case assumption holds. The multiplier $\lambda_1 = 2 > 0$ arrives at the end.

Surviving KKT point: $(x^*, y^*, \lambda_1^*, \lambda_2^*) = (1, 2, 2, 0)$, with $\mu_x^* = \mu_y^* = 0$.

Only the second binds

The case is $x + y < 3$ and $x + 2y = 6$, so $\lambda_1 = 0$ and $\lambda_2 \geq 0$. With $x, y > 0$, stationarity gives

$$x = 2 - \frac{\lambda_2}{2}, \quad y = 3 - \lambda_2,$$

and $x + 2y = 6$ turns into

$$8 - \frac{5}{2}\lambda_2 = 6 \Rightarrow (x, y, \lambda_2) = \left(\frac{8}{5}, \frac{11}{5}, \frac{4}{5}\right).$$

This candidate has $x + y = 19/5 > 3$, so it fails the case assumption and feasibility alike.

The same picture with the KKT point

- ◆ x^* is the point where the highest attainable level circle touches the feasible set.
- ◆ $\nabla f(x^*)$ points along the normal of the binding constraint, so $\nabla f(x^*) = \lambda_1^* \nabla g_1$.
- ◆ The dashed constraint is slack there, which is why $\lambda_2^* = 0$.

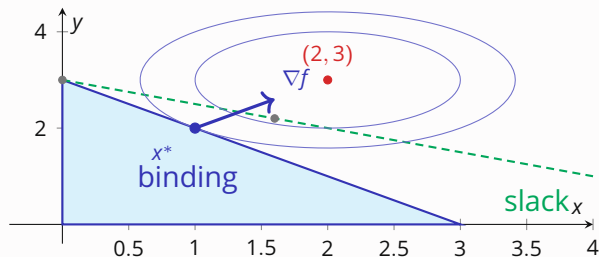


Figure 25 again, now with the solution of the case analysis: $(x^*, y^*) = (1, 2)$ and $(\lambda_1^*, \lambda_2^*) = (2, 0)$, the objective gradient at x^* , and the binding and slack constraints marked.

A curved inequality constraint

Every KKT step with a nonlinear g

For $\max x + y$ s.t. $x^2 + y^2 \leq 2$, write $\mathcal{L} = x + y + \lambda(2 - x^2 - y^2)$, $\lambda \geq 0$.

Stationarity. $1 - 2\lambda x = 0$ and $1 - 2\lambda y = 0$. At $\lambda = 0$ both give $1 = 0$, so $\lambda > 0$ and $x = y = 1/(2\lambda) > 0$.

Complementary slackness. $\lambda > 0$ forces $x^2 + y^2 = 2$, and $x = y > 0$ gives $2x^2 = 2$:

$$(x^*, y^*) = (1, 1), \quad \lambda^* = \frac{1}{2}, \quad x^* + y^* = 2.$$

Sufficiency. The objective is linear, hence concave, and g is convex, so this KKT point is the global maximizer.

Curvature. A linear constraint has one normal for its whole boundary; here $\nabla g = (2x, 2y)$ turns along the circle, and the KKT equations pick the boundary point whose normal $(2, 2)$ is parallel to ∇f .

KKT: a zero multiplier and a corner

Write the KKT conditions and solve each problem. Identify every binding constraint and multiplier.

(a) $\max_{x,y \geq 0} 4x + 2y - x^2 - y^2 \quad \text{s.t.} \quad x + y \leq 3.$

(b) $\max_{x,y \geq 0} 3x + y \quad \text{s.t.} \quad x + 2y \leq 6.$

A binding constraint with a zero multiplier

With $\mathcal{L} = 4x + 2y - x^2 - y^2 + \lambda(3 - x - y) + \mu x + \nu y$ and $\lambda, \mu, \nu \geq 0$, the conditions are

$$4 - 2x - \lambda + \mu = 0, \quad 2 - 2y - \lambda + \nu = 0, \quad \lambda(3 - x - y) = \mu x = \nu y = 0.$$

On $x, y > 0$ complementary slackness gives $\mu = \nu = 0$, so

$$x = 2 - \frac{\lambda}{2}, \quad y = 1 - \frac{\lambda}{2}, \quad 3 - x - y = \lambda.$$

The slack in the resource constraint equals λ , so $\lambda(3 - x - y) = \lambda^2 = 0$ and therefore $\lambda = 0$, giving $(x^*, y^*) = (2, 1)$ with $x^* + y^* = 3$.

The resource constraint binds at $(2, 1)$ and its multiplier is zero, the case that the converse implication would miss. Boundary candidates fail: at $x = y = 0$ the constraint is slack, so $\lambda = 0$ and the x equation asks for $\mu = -4$; at $x = 0 < y$, $\mu = \lambda - 4 \geq 0$ forces $\lambda \geq 4$ and then $y = 1 - \lambda/2 < 0$; at $y = 0 < x$, $\nu = \lambda - 2 \geq 0$ forces $\lambda \geq 2$, so the constraint binds at $x = 3$, while stationarity gives $x = 2 - \lambda/2 \leq 1$. Strict concavity with affine constraints makes $(2, 1)$ the unique global maximizer.

The conditions produce the corner

With $\mathcal{L} = 3x + y + \lambda(6 - x - 2y) + \mu x + \nu y$ and $\lambda, \mu, \nu \geq 0$,

$$3 - \lambda + \mu = 0, \quad 1 - 2\lambda + \nu = 0, \quad \lambda(6 - x - 2y) = \mu x = \nu y = 0.$$

The first equation gives $\lambda = 3 + \mu \geq 3 > 0$, so the resource constraint binds: $x + 2y = 6$. The second gives $\nu = 2\lambda - 1 \geq 5 > 0$, so $\nu y = 0$ forces $y = 0$. The binding constraint then gives $x = 6$, and $x > 0$ gives $\mu = 0$, hence $\lambda = 3$ and $\nu = 5$:

$$(x^*, y^*) = (6, 0), \quad \lambda^* = 3, \quad \mu^* = 0, \quad \nu^* = 5, \quad 3x^* + y^* = 18.$$

The corner comes out of the conditions themselves: $\lambda \geq 3$ forces the resource constraint to bind and $\nu > 0$ forces $y = 0$. Behind the algebra is the payoff per unit of the resource, 3 for x against 1/2 for y . The objective is linear and the constraints affine, so the KKT conditions are sufficient for a global maximum.

After the case analysis

The cases return the pairs (x, λ) that solve the KKT system. There can be none, one, or many, and two pairs can carry the same x .

None. Check existence, the constraint qualification, and the algebra.

One. Still a candidate. Concave f with convex g_j makes it a global maximizer.

Many. Each x is then a global maximizer and all attain the same value. Strict concavity of f gives a single x , while λ can still form a segment.

Case analysis finds KKT points; choosing the optimizer comes after.

KKT: necessity and sufficiency

Decide whether each statement is true or false and justify it.

- (a) At a differentiable local optimum, a constraint qualification makes KKT necessary.
- (b) For a concave maximization problem with convex inequality functions, any feasible KKT point is a global maximizer.
- (c) Any feasible point satisfying KKT is a global maximizer, regardless of curvature.

KKT: necessity and sufficiency

- (a) True. Differentiability plus a suitable constraint qualification gives the KKT necessary conditions at a local optimum.
- (b) True for the sign convention and hypotheses stated in this section: concavity of the objective and convexity of the functions defining the lower contour sets turn stationarity into $f(y) \leq f(x^*)$ for every feasible y .
- (c) False. For example, consider $\max_{-1 \leq x \leq 1} x^2$. At $x^* = 0$, both inequality constraints are slack, so their multipliers are zero, and stationarity holds because $f'(0) = 0$. Thus $x^* = 0$ satisfies KKT, but it is the global minimum; the global maximizers are $x = \pm 1$. Sufficiency of KKT needs the curvature assumptions of (b).

KKT: two binding constraints, and a minimization

Write the KKT conditions, work through the active sets, and report every multiplier.

- (a) $\max_{x,y \geq 0} xy$ s.t. $x + y \leq 6$, $x \leq 2$. Check the four active sets of the two resource constraints, then bound xy on the feasible set to show the KKT point is the global maximizer.
- (b) $\min_{x,y \geq 0} x^2 + y^2$ s.t. $x + y \geq 2$. Use the minimization convention: write the constraint as $g(x,y) = x + y - 2 \geq 0$, form $\mathcal{L} = f - \lambda g$, and report the sign of each multiplier.

KKT: two binding constraints, and a minimization

(a) The point $(1, 1)$ is feasible with $xy = 1 > 0$, while every feasible point with $x = 0$ or $y = 0$ has $xy = 0$. A maximizer therefore has $x, y > 0$, so the multipliers on $x \geq 0$ and $y \geq 0$ are zero. With $\mathcal{L} = xy + \lambda_1(6 - x - y) + \lambda_2(2 - x) + \mu_x x + \mu_y y$, the stationarity conditions give

$$y = \lambda_1 + \lambda_2, \quad x = \lambda_1.$$

NEITHER BINDS

$x + y < 6$ and $x < 2$ give $\lambda_1 = \lambda_2 = 0$, hence $x = y = 0$.

ONLY $x + y \leq 6$ BINDS

$x < 2$ gives $\lambda_2 = 0$, so $y = \lambda_1 = x$; with $x + y = 6$ this gives $x = y = 3$, outside $x < 2$.

ONLY $x \leq 2$ BINDS

$x + y < 6$ gives $\lambda_1 = 0$, so $x = \lambda_1 = 0$, against $x = 2$.

BOTH BIND

$x + y = 6$ and $x = 2$ give $y = 4$; then $x = \lambda_1$ gives $\lambda_1 = 2$ and $y = \lambda_1 + \lambda_2$ gives $\lambda_2 = 2$.

$(x^*, y^*) = (2, 4)$ with $\lambda_1^* = \lambda_2^* = 2$ and $x^*y^* = 8$. Feasibility gives $x \leq 2$ and $y \leq 6 - x$, so $xy \leq x(6 - x)$, which rises on $[0, 2]$; hence $xy \leq 8$ on the feasible set, with equality at $(2, 4)$.

KKT: two binding constraints, and a minimization

(b) Write $g(x, y) = x + y - 2 \geq 0$ and $\mathcal{L} = x^2 + y^2 - \lambda(x + y - 2)$ with $\lambda \geq 0$. Stationarity is $2x - \lambda = 0$ and $2y - \lambda = 0$, so $x = y = \lambda/2$. At $\lambda = 0$ the point $(0, 0)$ is infeasible, so $\lambda > 0$, the constraint binds, and

$$(x^*, y^*) = (1, 1), \quad \lambda^* = 2, \quad x^{*2} + y^{*2} = 2.$$

The two nonnegativity constraints are slack with zero multipliers. The objective is strictly convex and every constraint is affine, so by the KKT sufficiency page $(1, 1)$ is the unique global minimizer. The sign restriction $\lambda \geq 0$ accompanies the nonnegative slack $g \geq 0$; in the minimization convention the multiplier enters as $-\lambda g$. The standard form of the earlier page writes $\tilde{g} = -g \leq 0$, and $-\lambda g = +\lambda \tilde{g}$ is the same Lagrangian.

An equality and an inequality in one problem

Two kinds of multiplier

For a parameter $c \in \mathbb{R}$, consider

$$\min_{x,y} x^2 + y^2 \quad \text{s.t.} \quad x + y = 2, \quad x \geq c,$$

written in the minimization convention as $h(x,y) = 2 - x - y = 0$, $g(x,y) = x - c \geq 0$, with $\mathcal{L} = x^2 + y^2 - \lambda g - \nu h$, $\lambda \geq 0$, $\nu \in \mathbb{R}$.

- Solve the KKT system at $c = 0$ and at $c = 3/2$.
- Report which constraint binds in each case, and the sign of ν^* .
- Say why the sign restriction applies to λ^* and why ν^* is free in sign.
- Find $V(c)$ on both sides of $c = 1$ and check $V'(c) = \lambda^*(c)$.

An equality and an inequality in one problem

The equality multiplier is negative here

Stationarity gives $2x - \lambda + \nu = 0$ and $2y + \nu = 0$, so $\nu = -2y$ and $\lambda = 2x + \nu = 2x - 2y$.

(a) At $c = 0$: the guess $\lambda = 0$ gives $x = y$, and $x + y = 2$ gives $(1, 1)$, where $g(1, 1) = 1 > 0$ and $\lambda^* = 0$, so complementary slackness holds; $\nu^* = -2$. At $c = 3/2$: $\lambda = 0$ would give $(1, 1)$, which violates $x \geq 3/2$, so $\lambda > 0$, $x^* = 3/2$, $y^* = 1/2$, $\nu^* = -1$ and $\lambda^* = 2$; both are unique global minimizers (strict convexity, affine constraints).

(b) At $c = 0$ the equality binds and $x \geq c$ is slack with $\lambda^* = 0$; at $c = 3/2$ both bind. In each case $\nu^* < 0$.

(c) An equality constraint restricts movement in both directions, so its multiplier takes either sign; $x \geq c$ restricts one direction, and $\lambda \geq 0$ records that relaxing it can raise the feasible set only on that side.

(d) For $c \leq 1$, $V(c) = 2$ and $\lambda^* = 0$. For $c > 1$, $x^* = c$, $y^* = 2 - c$ and $V(c) = 2c^2 - 4c + 4$, so $V'(c) = 4c - 4$, which equals $\lambda^* = 2x^* - 2y^* = 4c - 4$. Both expressions give $V'(1) = 0$, so V is differentiable at the switch.

Section 5

Comparative statics

Parameterized problems: solution and value

For a parameter θ , let the feasible set be $S(\theta)$ and the objective be $f(x, \theta)$. The problem defines a maximizer set and a value,

$$C(\theta) = \arg \max_{x \in S(\theta)} f(x, \theta), \quad V(\theta) = \sup_{x \in S(\theta)} f(x, \theta),$$

the **solution correspondence**, which **may contain several maximizers**, and the **value function**, which equals $f(x^*, \theta)$ at every $x^* \in C(\theta)$ when the maximum is **attained**.

Two different derivatives. Choice sensitivity $dx^*/d\theta$ is comparative statics, from the implicit function theorem; value sensitivity $dV/d\theta$ is the envelope theorem, which often avoids $dx^*/d\theta$.

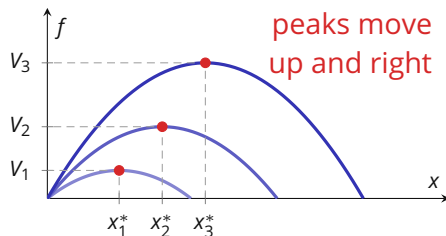


Figure 26 · Objectives $x \mapsto f(x, \theta)$ for $\theta_1 < \theta_2 < \theta_3$ (lighter to darker): the peaks' positions trace $C(\theta)$, their heights trace $V(\theta)$.

Video: [Matt Clancy, Comparative statics](#).

A solution correspondence can be set-valued

Selections from the maximizer set

Let $x \in \{0, 1\}$ and $f(x, \theta) = x(\theta - k)$ with k fixed. Then

$$C(\theta) = \begin{cases} \{0\}, & \theta < k, \\ \{0, 1\}, & \theta = k, \\ \{1\}, & \theta > k, \end{cases} \quad V(\theta) = \max\{0, \theta - k\}.$$

The maximizer set C is a correspondence: at $\theta = k$ the set $C(\theta)$ has two elements. A function $\theta \mapsto x^*(\theta)$ with $x^*(\theta) \in C(\theta)$ is a **selection** from C ; at $\theta = k$ both choices are available, $x^*(k) = 0$ or $x^*(k) = 1$. Where $C(\theta)$ is a singleton the selection is forced. Every selection jumps from 0 to 1 as θ crosses k ; the value function is continuous, with a kink at k .

The next pages differentiate $x^*(\theta)$ and $V(\theta)$ along a differentiable solution path. This example has **no such path at $\theta = k$** : $C(k)$ has two elements and every selection jumps there.

A set-valued solution correspondence

A maximizer set with two elements

Let $S = [-2, 2]$ and $f(x, \theta) = -(x^2 - 1)^2 + \theta x$ with $\theta \in \mathbb{R}$.

- (a) Find $C(0)$ and $V(0)$.
- (b) Compare $f(x, \theta)$ with $f(-x, \theta)$ and deduce that every maximizer is positive for $\theta > 0$ and negative for $\theta < 0$.

A set-valued solution correspondence

Symmetry fixes the sign of every maximizer

- (a) At $\theta = 0$ the objective is $-(x^2 - 1)^2$, which is 0 at $x = \pm 1$ and negative elsewhere:
 $C(0) = \{-1, 1\}$ and $V(0) = 0$.
- (b) The first term is even in x , so $f(x, \theta) - f(-x, \theta) = 2\theta x$. For $\theta > 0$ every $x > 0$ beats its mirror image $-x$, and $f(1, \theta) = \theta > -1 = f(0, \theta)$, so every maximizer is positive; for $\theta < 0$ the inequality reverses.

Consumer demand and indirect utility

A consumer with prices $p \gg 0$ and wealth w has

$$B(p, w) = \{x \in \mathbb{R}_+^n : p \cdot x \leq w\},$$

$$C(p, w) = \arg \max_{x \in B(p, w)} u(x),$$

$$V(p, w) = \sup_{x \in B(p, w)} u(x).$$

If u is strictly quasiconcave on the convex budget set, $C(p, w)$ is a singleton whenever it is nonempty and can be written as an ordinary demand function. $C(p, w)$ is a set of bundles and $V(p, w)$ is a number.

Continuity properties of C and V belong to the later optimization course.

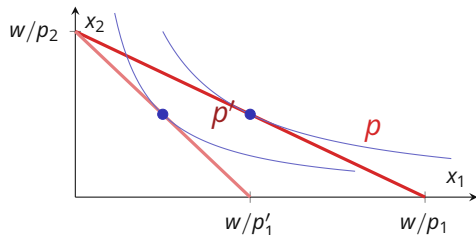


Figure 27 · A higher price p'_1 rotates the budget line inward; the optimum moves from the right dot, $C(p, w)$, to the left dot, $C(p', w)$, and the indifference level reached, V , falls.

Firm supply from an FOC

Interior output choice

A price-taking firm chooses an interior output $q^*(p)$ in $\max_{q>0} pq - c(q)$, with $c''(q) > 0$. The first-order condition $p - c'(q^*(p)) = 0$ holds at every price, so it is an identity in p .

Differentiate the identity

$$1 - c''(q^*(p)) \frac{dq^*}{dp} = 0, \quad \boxed{\frac{dq^*}{dp} = \frac{1}{c''(q^*(p))} > 0.}$$

This is the implicit function theorem from Differentiation applied to an optimization FOC: with strictly convex cost, a higher output price raises the firm's optimal output locally.

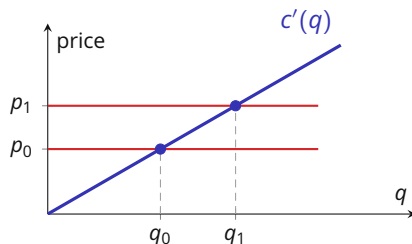


Figure 28 · Marginal cost $c'(q)$ and two price lines. The FOC $p = c'(q^*)$ is the crossing, a higher price moves it right, and the slope of c' at q^* is c'' .

Differentiate an FOC

Let $\theta > 0$ and consider

$$\max_{q>0} \left\{ \theta q - \frac{1}{3}q^3 \right\}.$$

- (a) Find $q^*(\theta)$ and differentiate it directly.
- (b) Differentiate the FOC implicitly, as for the firm's supply above, and verify the same derivative.

Differentiate an FOC

The FOC is $\theta - q^2 = 0$, and the objective is strictly concave on $q > 0$. Thus

$$q^*(\theta) = \sqrt{\theta}, \quad \frac{dq^*}{d\theta} = \frac{1}{2\sqrt{\theta}}.$$

Implicit differentiation of the identity $\theta - q^*(\theta)^2 = 0$ gives $1 - 2q^* \frac{dq^*}{d\theta} = 0$, hence

$$\frac{dq^*}{d\theta} = \frac{1}{2q^*(\theta)} = \frac{1}{2\sqrt{\theta}},$$

the same derivative, obtained without first solving for q^* .

Differentiation applied: local derivative of an optimizer

Differentiate the first-order condition

Suppose F is continuously differentiable near a point where an interior optimizer satisfies

$$F(x^*(\theta), \theta) = 0,$$

and $D_x F$ is **invertible** there. The implicit function theorem gives

$$\frac{dx^*}{d\theta} = -[D_x F(x^*, \theta)]^{-1} D_\theta F(x^*, \theta).$$

For $F = \nabla_x f$, continuous differentiability of F means f is twice continuously differentiable, and the matrix $D_x F$ is the Hessian with respect to x . The Jacobian comes from Differentiation; solving for $dx^*/d\theta$ uses matrix invertibility from Linear Algebra.

Further reading: [Boyd, Ch. 15](#); §19.3; [Border, Implicit Function Theorem](#).

Video: [Arizona Math Camp 37\(A\)](#), the implicit function theorem.

Matrix comparative statics

Comparative statics in matrix form

Let

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}, \quad b(\theta) = \begin{bmatrix} 5 + \theta \\ 4 - \theta \end{bmatrix},$$

and consider

$$\max_{x \in \mathbb{R}^2} \left\{ b(\theta)^\top x - \frac{1}{2} x^\top A x \right\}.$$

Find $x^*(\theta)$ and $dx^*/d\theta$. Then obtain the same derivative by differentiating the FOC $Ax^*(\theta) = b(\theta)$.

Matrix comparative statics

Two routes to the same derivative

Since $A^{-1} = \frac{1}{3} \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$,

$$x^*(\theta) = A^{-1}b(\theta) = \begin{bmatrix} 2 + \theta \\ 1 - \theta \end{bmatrix}, \quad \frac{dx^*}{d\theta} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

Differentiating the FOC gives

$$A \frac{dx^*}{d\theta} = \frac{db}{d\theta} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \quad \frac{dx^*}{d\theta} = A^{-1} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

This is the formula of the preceding page with $F = b(\theta) - Ax$: $D_x F = -A$ and $D_\theta F = b'(\theta)$. Because $-A \prec 0$, this critical point is the unique global maximizer for every θ .

Section 6

Envelope theorem

Differentiation applied: why the indirect effect disappears

Let an interior optimizer $x^*(\alpha)$, differentiable in α , define

$$V(\alpha) = \max_x f(x, \alpha) = f(x^*(\alpha), \alpha).$$

The chain rule gives

$$V'(\alpha) = f_x(x^*(\alpha), \alpha) \frac{dx^*}{d\alpha} + f_\alpha(x^*(\alpha), \alpha).$$

At an **interior optimum**, the FOC is

$$f_x(x^*(\alpha), \alpha) = 0.$$

Therefore

$$\begin{aligned} V'(\alpha) &= 0 \cdot \frac{dx^*}{d\alpha} + f_\alpha(x^*(\alpha), \alpha) \\ &= \boxed{f_\alpha(x^*(\alpha), \alpha)}. \end{aligned}$$

This is the chain rule from Differentiation applied at a point where the FOC makes the indirect first-order term zero.

Video: [Arizona Math Camp 38\(A\)](#), the envelope theorem.

Two derivatives from one parameterized problem

The name: for each fixed choice $\bar{x}$, the map $\alpha \mapsto f(\bar{x}, \alpha)$ is one curve, and V is the upper envelope of these curves. At α_0 , V and the curve of the current optimizer, $\alpha \mapsto f(x^*(\alpha_0), \alpha)$, share the same value and the same slope, since $V(\alpha_0) = f(x^*(\alpha_0), \alpha_0)$ and $V'(\alpha_0) = f_\alpha(x^*(\alpha_0), \alpha_0)$. That tangency is the geometric side of the envelope theorem, because the indirect term $f_x(x^*, \alpha) x^{*'}(\alpha)$ vanishes through $f_x(x^*, \alpha) = 0$.

Further reading: [Boyd, §19.1](#). Video: [Arizona Math Camp, The Envelope Theorem and Shadow Values](#).

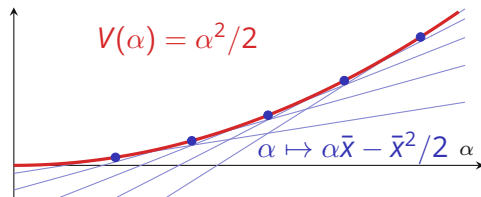


Figure 29 · $f(x, \alpha) = \alpha x - x^2/2$: for fixed $\bar{x}$ the map $\alpha \mapsto \alpha\bar{x} - \bar{x}^2/2$ is a line; $V(\alpha) = \alpha^2/2$ is their upper envelope, tangent to the line of $\bar{x} = \alpha$.

Equality-constrained derivation

Consider

$$V(\alpha) = \max_x f(x, \alpha) \quad \text{s.t.} \quad g(x, \alpha) = 0, \quad \mathcal{L} = f + \lambda(0 - g).$$

Along a differentiable solution path satisfying the Lagrange equations,

$$V' = f_x x^{*'} + f_\alpha.$$

The Lagrange FOC and the derivative of the binding constraint give

$$f_x = \lambda^* g_x, \quad g_x x^{*'} + g_\alpha = 0 \quad \Rightarrow \quad g_x x^{*'} = -g_\alpha.$$

Substituting,

$$V' = \lambda^* g_x x^{*'} + f_\alpha = f_\alpha - \lambda^* g_\alpha = \boxed{\mathcal{L}_\alpha(x^*, \lambda^*, \alpha)}.$$

The direct parameter effect remains, together with the parameter's effect on the binding constraint at the shadow price.

Video: [Arizona Math Camp 38\(B\), constrained envelope](#).

Shadow prices

With the sign convention $\mathcal{L} = f + \sum_j \lambda_j(b_j - g_j)$, if the value function V is differentiable, the envelope theorem gives

$$\lambda_j^* = \frac{\partial V}{\partial b_j}.$$

Thus the multiplier is the local marginal value of increasing the corresponding right-hand side b_j ; when larger b_j means more resources, this is the value of relaxing the constraint.

- ◆ Consumer theory: λ is the marginal utility of income.
- ◆ Firm theory: λ is the marginal value of relaxing a resource constraint.

The previous page derives this identity.

Video: [Arizona Math Camp 39\(A\), shadow values.](#)

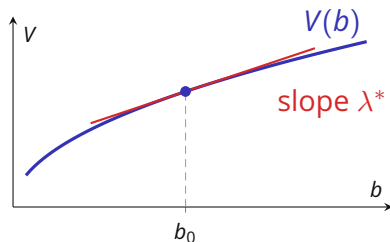


Figure 30 · The value function $V(b)$ and its tangent at b_0 :
the slope is the multiplier λ^* .

Inequality-constrained envelope

Consider a KKT problem in which the objective, constraints, and right-hand sides are differentiable functions of a scalar parameter α :

$$V(\alpha) = \max_x f(x, \alpha) \quad \text{s.t.} \quad g_j(x, \alpha) \leq b_j(\alpha).$$

With

$$\mathcal{L} = f + \sum_j \lambda_j (b_j - g_j),$$

along a differentiable solution with a **locally unchanged active set**,

$$V'(\alpha) = \mathcal{L}_\alpha(x^*, \lambda^*, \alpha) = f_\alpha(x^*, \alpha) + \sum_j \lambda_j^* (b'_j(\alpha) - g_{j,\alpha}(x^*, \alpha)).$$

- ◆ Parameter in the **objective**: if only f depends on α , then $V'(\alpha) = f_\alpha(x^*, \alpha)$.
- ◆ Parameter in the **right-hand side**: with $\alpha = b_j$, $\partial V / \partial b_j = \lambda_j^*$, the shadow price.
- ◆ Parameter in the **constraint function**: the effect is the term $-\lambda_j^* g_{j,\alpha}$, as in the budget exercise where $g(x, y, p) = px + y$.

Complementary slackness and shadow prices

With $V(b) = \max_x \{f(x) : g_j(x) \leq b_j\}$ and $\mathcal{L} = f + \sum_j \lambda_j(b_j - g_j)$, along a differentiable KKT solution path with a **locally unchanged active set**, $\partial V / \partial b_j = \lambda_j^*$. Together with complementary slackness:

- ◆ $g_j(x^*) < b_j \Rightarrow \lambda_j^* = 0$, so $\partial V / \partial b_j = 0$: relaxing an unused constraint is worth nothing.
- ◆ $\lambda_j^* > 0 \Rightarrow g_j(x^*) = b_j$: a constraint can have a positive shadow price only if it binds.
- ◆ **A binding constraint still allows $\lambda_j^* = 0$** , so binding by itself leaves the shadow price free to vanish.

In the KKT example with two resource constraints, $\lambda_1^* = 2$ and $\lambda_2^* = 0$: relaxing the first constraint is worth 2, the slack second one nothing.

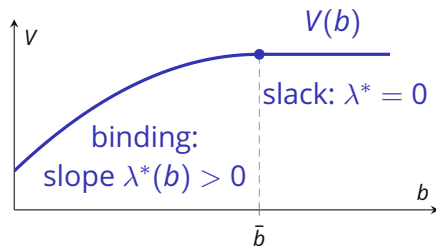


Figure 31 · One constraint, one right-hand side b : V rises with slope $\lambda^*(b) > 0$ while the constraint binds and is flat once it is slack; the active set changes at $\bar{b}$.

A binding constraint with a zero shadow price

The first KKT exercise had $(x^*, y^*) = (2, 1)$ with $x^* + y^* = 3$ and $\lambda^* = 0$: the resource constraint binds and its multiplier vanishes. Make the right-hand side a parameter,

$$V(b) = \max_{x, y \geq 0} 4x + 2y - x^2 - y^2 \quad \text{s.t.} \quad x + y \leq b.$$

- (a) For $1 \leq b \leq 3$, solve for $x^*(b)$, $y^*(b)$, $\lambda^*(b)$ and $V(b)$.
- (b) For $b \geq 3$, show that the unconstrained maximizer is feasible and find $V(b)$.
- (c) Check $V'(b) = \lambda^*(b)$ on both sides of $b = 3$, and say whether V has a kink there.
- (d) Explain in one sentence how a constraint can bind and still have zero marginal value.

A binding constraint with a zero shadow price

(a) Write $\mathcal{L} = 4x + 2y - x^2 - y^2 + \lambda(b - x - y) + \mu_x x + \mu_y y$, so that

$$4 - 2x - \lambda + \mu_x = 0, \quad 2 - 2y - \lambda + \mu_y = 0, \quad \lambda(b - x - y) = \mu_x x = \mu_y y = 0.$$

Suppose the resource constraint is slack. Then $\lambda = 0$, so $\mu_x = 2x - 4$ and $\mu_y = 2y - 2$; at $x = 0$ this asks for $\mu_x = -4$, so $x > 0$, $\mu_x = 0$ and $x = 2$, and the same steps give $y = 1$. That candidate has $x + y = 3$, which stays inside the constraint only when $b > 3$. Hence the constraint binds for $b < 3$.

With $x + y = b$ and $x, y > 0$, stationarity gives $4 - 2x = \lambda$ and $2 - 2y = \lambda$, so $x - y = 1$ and

$$x^* = \frac{b+1}{2}, \quad y^* = \frac{b-1}{2}, \quad \lambda^* = 3 - b > 0, \quad V(b) = 3b + 1 - \frac{1}{2}(b^2 + 1).$$

The condition $b \geq 1$ keeps $y^* \geq 0$, and at $b = 1$ the maximum is attained at the boundary, with $y^* = 0$ and $\mu_y = 2y^* - 2 + \lambda^* = 0$: the constraint $y \geq 0$ binds there with a zero multiplier. The objective is strictly concave and the constraints affine, so the KKT point is the global maximizer.

(b) The unconstrained maximizer of $4x + 2y - x^2 - y^2$ is $(2, 1)$, which satisfies $x + y = 3 \leq b$. So $V(b) = 5$ for every $b \geq 3$, with $\lambda^* = 0$.

A binding constraint with a zero shadow price

- (c) For $b < 3$, $V'(b) = 3 - b = \lambda^*(b)$; for $b > 3$, $V'(b) = 0 = \lambda^*(b)$. Both the left and right derivatives at $b = 3$ are 0, so V has no kink where the multiplier reaches zero.
- (d) At $b = 3$ the constraint holds with equality at the point the problem would choose anyway, so a small increase in b has no effect on the optimized value.

Envelope theorem: direct calculation and a shadow price

- (a) Let $V(\alpha) = \max_{x \in \mathbb{R}} \{\alpha x - \frac{1}{2}x^2\}$. Find $x^*(\alpha)$, compute $V'(\alpha)$ directly, and recover it from the envelope theorem.
- (b) For $b > 0$, let $V(b) = \max_{x,y>0} \{\ln x + \ln y : x + y = b\}$. Find $\lambda^*(b)$ and verify $V'(b) = \lambda^*(b)$.

Envelope theorem: direct calculation and a shadow price

For (a), the FOC gives $x^*(\alpha) = \alpha$, so

$$V(\alpha) = \frac{1}{2}\alpha^2, \quad V'(\alpha) = \alpha.$$

The envelope derivative is $f_\alpha(x^*, \alpha) = x^* = \alpha$. For (b), with $\mathcal{L} = \ln x + \ln y + \lambda(b - x - y)$, the FOCs give

$$x^* = y^* = \frac{b}{2}, \quad \lambda^* = \frac{2}{b}.$$

Strict concavity of $\ln x + \ln y$ with the affine constraint makes this the unique global maximizer. Since $V(b) = 2 \ln(b/2)$, direct differentiation gives $V'(b) = 2/b = \lambda^*(b)$.

Envelope theorem: a parameter inside the constraint

For $p, m > 0$, let

$$V(p) = \max_{x,y>0} \{\ln x + \ln y : px + y \leq m\}.$$

- (a) Solve for x^* , y^* , and λ^* .
- (b) Use the envelope theorem to find V_p . Which term of $\mathcal{L}_\alpha$ on the preceding pages survives here?
- (c) Verify the answer by differentiating $V(p)$ directly.
- (d) Compute dx^*/dp and dy^*/dp and interpret the signs. Which of the two sensitivity questions does each of (b) and (d) answer?

Envelope theorem: a parameter inside the constraint

The objective is increasing in both goods, so the budget binds. With $\mathcal{L} = \ln x + \ln y + \lambda(m - px - y)$, the FOCs are

$$\frac{1}{x} = \lambda p, \quad \frac{1}{y} = \lambda, \quad \text{so} \quad px = \frac{1}{\lambda}, \quad y = \frac{1}{\lambda},$$

and $px + y = 2/\lambda = m$ gives

$$\lambda^* = \frac{2}{m}, \quad x^* = \frac{m}{2p}, \quad y^* = \frac{m}{2}.$$

The objective is strictly concave and the budget constraint affine, so this KKT point is the unique global maximizer. Here p enters the constraint function $g(x, y, p) = px + y$, with $g_p = x$, while the objective and the right-hand side m are free of p . Only the term $-\lambda^* g_p$ of $\mathcal{L}_\alpha$ survives:

$$V_p = -\lambda^* x^* = -\frac{2}{m} \cdot \frac{m}{2p} = -\frac{1}{p}.$$

Directly, $V(p) = \ln \frac{m}{2p} + \ln \frac{m}{2} = 2 \ln \frac{m}{2} - \ln p$, so $V_p = -1/p$. Roy's identity below states this relation for a general utility function.

Envelope theorem: the optimizer as well as the value

(d) The solution of (a) gives the two derivatives directly:

$$\frac{dx^*}{dp} = \frac{d}{dp} \frac{m}{2p} = -\frac{m}{2p^2} < 0, \quad \frac{dy^*}{dp} = \frac{d}{dp} \frac{m}{2} = 0.$$

A higher price of the first good shrinks the feasible set in the x direction, and the demand for that good falls. With the log objective the expenditure on each good is half of wealth at every price, so $px^* = m/2$ and the quantity y^* stays put.

Here the parameter enters the feasible set through the constraint function $g(x, y, p) = px + y$, and the budget line pivots as p changes.

Part (b) answers the second sensitivity question, how the optimized value moves:

$V_p = -\lambda^* x^* = -1/p$; part (d) answers the first, how the optimizer moves. The envelope theorem gives the value derivative without dx^*/dp .

Envelope theorem: both terms at once

Each earlier exercise switched on one term of $\mathcal{L}_\alpha$. For $\alpha > 0$, let

$$V(\alpha) = \max_{x>0} \{ \alpha \ln x : x \leq 1 + \alpha \},$$

where the parameter multiplies the objective and also shifts the right-hand side.

- (a) Solve for $x^*(\alpha)$ and $\lambda^*(\alpha)$.
- (b) Write $\mathcal{L}_\alpha$ and identify the two terms f_α and $\lambda^*[b'(\alpha) - g_\alpha]$.
- (c) Compute $V'(\alpha)$ with the envelope theorem and check it by differentiating V directly.

Envelope theorem: both terms at once

(a) The objective rises in x for $\alpha > 0$, so the constraint binds: $x^*(\alpha) = 1 + \alpha$. With $\mathcal{L} = \alpha \ln x + \lambda(1 + \alpha - x)$, stationarity gives $\alpha/x^* = \lambda^*$, so $\lambda^*(\alpha) = \alpha/(1 + \alpha) > 0$.

(b) Here $f(x, \alpha) = \alpha \ln x$ gives $f_\alpha = \ln x^* = \ln(1 + \alpha)$, and $b(\alpha) = 1 + \alpha$ with $g(x) = x$ gives $b'(\alpha) - g_\alpha = 1 - 0 = 1$. Both terms are nonzero:

$$\mathcal{L}_\alpha = \ln(1 + \alpha) + \lambda^* \cdot 1.$$

(c) The envelope theorem gives $V'(\alpha) = \ln(1 + \alpha) + \alpha/(1 + \alpha)$. Directly, $V(\alpha) = \alpha \ln(1 + \alpha)$, so

$$V'(\alpha) = \ln(1 + \alpha) + \frac{\alpha}{1 + \alpha},$$

the same expression: the first term is the parameter acting through the objective, the second is the parameter acting through the constraint.

A binary optimizer jump and a kink in the value function

The binary choice again

Return to $x \in \{0, 1\}$ and $f(x, \theta) = x(\theta - k)$, so that $V(\theta) = \max\{0, \theta - k\}$. Away from $\theta = k$ the optimizer is constant on each side of k and the envelope formula holds:

$$V'(\theta) = f_{\theta}(x^*(\theta), \theta) = x^*(\theta) = \begin{cases} 0, & \theta < k, \\ 1, & \theta > k. \end{cases}$$

At $\theta = k$ the optimizer jumps from 0 to 1, the slopes $f_{\theta}(0, \theta) = 0$ and $f_{\theta}(1, \theta) = 1$ of the two curves differ, and V has no derivative. The kink comes from the differing slopes: a jump between curves whose slopes agree at the crossing keeps V differentiable there. Away from the jump the optimizer is constant, one case of the differentiable solution path the preceding pages assumed; at the jump there is no such path. The left and right derivatives are still the envelope values of the two competing choices.

KKT, parameter changes & envelope

Optional cumulative exercise: KKT, a parameter, and the envelope theorem

For $\alpha \geq 0$, consider

$$\sup_{x,y>0} \alpha \ln x + \ln y$$

subject to

$$2x + y \leq 6.$$

- For $\alpha > 0$, write the KKT conditions and solve for $x^*(\alpha)$ and $y^*(\alpha)$.
- At $\alpha = 0$, determine whether a maximizer exists and find the supremum.
- Let $V(\alpha)$ be the supremum. Use the envelope theorem to determine where V decreases and increases.
- Add $x \leq y$. Determine whether the solution changes at $\alpha = 1$ and $\alpha = 3$; find the revised solution when it changes.

KKT, parameter changes and envelope

Part (a): KKT system and solution

With multiplier $\lambda \geq 0$,

$$\mathcal{L} = \alpha \ln x + \ln y + \lambda(6 - 2x - y).$$

The KKT conditions are

$$\frac{\alpha}{x} = 2\lambda, \quad \frac{1}{y} = \lambda, \quad \lambda \geq 0, \quad 2x + y \leq 6, \quad \lambda(6 - 2x - y) = 0.$$

The objective is increasing in both variables, so the resource constraint binds: $2x + y = 6$. The first-order conditions imply

$$x = \frac{\alpha}{2\lambda}, \quad y = \frac{1}{\lambda}, \quad 2x + y = \frac{\alpha + 1}{\lambda} = 6.$$

Hence

$$x^*(\alpha) = \frac{3\alpha}{\alpha + 1}, \quad y^*(\alpha) = \frac{6}{\alpha + 1}, \quad \lambda^*(\alpha) = \frac{\alpha + 1}{6}.$$

For $\alpha > 0$ strict concavity with an affine constraint makes this the unique global maximizer.

KKT, parameter changes and envelope

Parts (b) and (c): the supremum at $\alpha = 0$ and the envelope derivative

At $\alpha = 0$, the objective is $\ln y$. Feasibility with $x > 0$ implies $y < 6$, so no maximizer exists and

$$V(0) = \sup_{x, y > 0, 2x + y \leq 6} \ln y = \ln 6.$$

For $\alpha > 0$,

$$V(\alpha) = \ln 6 + \alpha \ln 3 + \alpha \ln \alpha - (\alpha + 1) \ln(\alpha + 1).$$

The envelope theorem gives

$$V'(\alpha) = \ln x^*(\alpha) = \ln \left(\frac{3\alpha}{\alpha + 1} \right).$$

Since $V(\alpha) \rightarrow \ln 6$ as $\alpha \downarrow 0$, V is continuous at zero. It decreases on $[0, 1/2]$ and increases on $[1/2, \infty)$.

KKT, parameter changes and envelope

Part (d): adding the constraint $x \leq y$

For $\alpha > 0$,

$$\frac{x^*(\alpha)}{y^*(\alpha)} = \frac{\alpha}{2}.$$

At $\alpha = 1$, $(x^*, y^*) = (3/2, 3)$, so $x \leq y$ is satisfied. At $\alpha = 3$, $(x^*, y^*) = (9/4, 3/2)$, so the added constraint changes the solution. The binding resource constraint gives $y = 6 - 2x$, and $x \leq y$ becomes $x \leq 2$, so the problem reduces to a single variable,

$$\max_{0 < x \leq 2} \varphi(x) = 3 \ln x + \ln(6 - 2x), \quad \varphi'(x) = \frac{3}{x} - \frac{2}{6 - 2x}.$$

Solving $\varphi'(x) = 0$ gives $x = 9/4 > 2$, and φ' decreases, so $\varphi' > 0$ throughout $(0, 2]$ and the maximum is attained at the right endpoint, $x^* = 2$ and $y^* = 6 - 2(2) = 2$.

Roy's identity

Roy's identity from the value function

Define the indirect utility function by

$$v(p, w) = \max_{x \geq 0} \{u(x) : p \cdot x \leq w\}.$$

If v is differentiable at (p, w) and $v_w(p, w) > 0$, Marshallian demand satisfies

$$x_i(p, w) = -\frac{v_{p_i}(p, w)}{v_w(p, w)}.$$

This is the envelope theorem applied to the consumer's value function.

Check with the Cobb-Douglas log utility of the worked example: $\tilde{v}(p, w) = \alpha \ln \frac{\alpha w}{(\alpha + \beta)p_1} + \beta \ln \frac{\beta w}{(\alpha + \beta)p_2}$ gives $\tilde{v}_{p_1} = -\alpha/p_1$ and $\tilde{v}_w = (\alpha + \beta)/w$, so $-\tilde{v}_{p_1}/\tilde{v}_w = \frac{\alpha}{\alpha + \beta} \frac{w}{p_1} = x_1^*$.

Video: [Economics in Many Lessons, Roy's identity](#).

Hotelling and Shephard

Envelope identities

Define the profit, cost and expenditure functions by

$$\pi(p) = \max_{y \in Y} p \cdot y,$$

$$c(w, q) = \min_{x \geq 0} \{w \cdot x : F(x) \geq q\},$$

$$e(p, \bar{u}) = \min_{x \geq 0} \{p \cdot x : u(x) \geq \bar{u}\}.$$

At price vectors where π and $c(\cdot, q)$ are differentiable:

Hotelling net supply from profit

$$\frac{\partial \pi(p)}{\partial p_i} = y_i(p),$$

Shephard conditional factor demand from cost

$$\frac{\partial c(w, q)}{\partial w_i} = x_i(w, q).$$

With $c(q) = q^2/2$, the envelope exercise gave $q^*(p) = p$ and $\pi(p) = p^2/2$, so $\pi'(p) = p = q^*(p)$: Hotelling in one line. With $F(x) = \min\{x_1, x_2\}$, the cost function is $c(w, q) = (w_1 + w_2)q$, so $\partial c/\partial w_1 = q = x_1(w, q)$: Shephard in one line.

Videos: [Arizona Math Camp](#), [Hotelling's lemma](#); [Shephard's lemma](#).

Roy, Hotelling and Shephard on one example each

Verify each identity on a value function in closed form

- (a) With $u(x) = x_1^{1/2}x_2^{1/2}$, the indirect utility is $v(p, w) = w/(2\sqrt{p_1p_2})$. Compute $-v_{p_1}/v_w$ and compare it with the Marshallian demand $x_1(p, w) = w/(2p_1)$.
- (b) With one output and cost $c(q) = q^3/3$, find $q^*(p)$ and $\pi(p)$, then check $\pi'(p) = q^*(p)$.
- (c) With $F(x) = x_1^{1/2}x_2^{1/2}$, find the cost function $c(w, q)$ and the conditional demand $x_1(w, q)$, then check $\partial c/\partial w_1 = x_1(w, q)$.

Roy, Hotelling and Shephard on one example each

Each identity holds by direct differentiation

(a) $v_{p_1} = -w/(4p_1^{3/2}p_2^{1/2})$ and $v_w = 1/(2\sqrt{p_1p_2})$, so

$$-\frac{v_{p_1}}{v_w} = \frac{w}{4p_1^{3/2}p_2^{1/2}} \cdot 2\sqrt{p_1p_2} = \frac{w}{2p_1} = x_1(p, w).$$

(b) Profit is $\max_q \{pq - q^3/3\}$, so $p = q^2$ and $q^*(p) = \sqrt{p}$. Then $\pi(p) = p^{3/2} - p^{3/2}/3 = \frac{2}{3}p^{3/2}$ and $\pi'(p) = p^{1/2} = q^*(p)$.

Shephard: the cost minimization behind the identity

Conditional demands and the cost function

(c) Minimize $w_1x_1 + w_2x_2$ subject to $\sqrt{x_1x_2} \geq q$. Since $w_1, w_2 > 0$, the production constraint binds, $x_1x_2 = q^2$, so $x_2 = q^2/x_1$ and the cost along the isoquant is $C(x_1) = w_1x_1 + w_2q^2/x_1$. Then $C'(x_1) = w_1 - w_2q^2/x_1^2 = 0$ gives $x_1^2 = q^2w_2/w_1$, and $x_1 > 0$ selects $x_1(w, q) = q\sqrt{w_2/w_1}$, hence $x_2(w, q) = q^2/x_1 = q\sqrt{w_1/w_2}$. Adding the two terms gives $c(w, q) = 2q\sqrt{w_1w_2}$ and

$$\frac{\partial c}{\partial w_1} = q\sqrt{\frac{w_2}{w_1}} = x_1(w, q).$$

In each case the derivative of the value function returns the optimal choice itself.

Convex duality behind the value functions

Expenditure, cost, and profit

These value functions inherit curvature from pointwise extrema of linear functions of prices:

$e(p, \bar{u})$ and $c(w, q)$ are concave and homogeneous of degree one in their price vectors.

$\pi(p)$ is convex and homogeneous of degree one in netput prices.

Shephard's and Hotelling's identities connect price derivatives to conditional demands and net supplies.

Study strategy. When you cannot memorize all these properties in Micro I, derive the curvature and homogeneity from the value function definitions.

Further reading: Boyd, Ch. 7, §§7.4.3–7.5.6 (pp. 29–39); Ch. 13, §13.3.5 (pp. 24–25).

Profit and cost as envelopes of linear functions

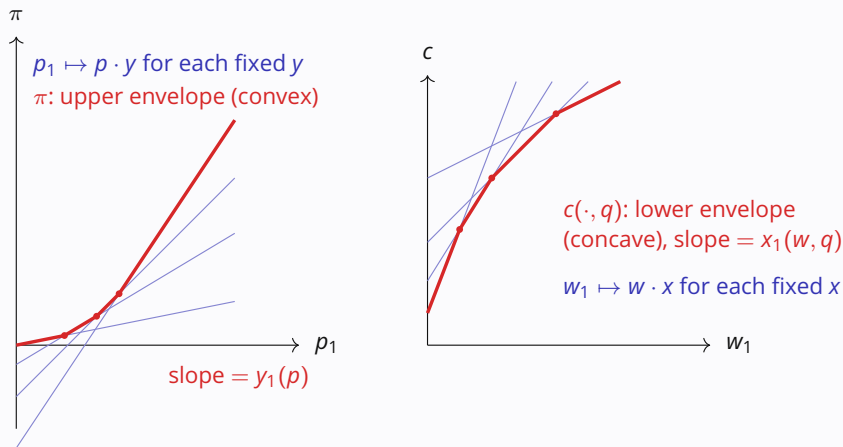


Figure 32 · Left: each fixed netput $y \in Y$ gives a line in p_1 ; π is their upper envelope, hence convex, with slope $y_1(p)$. Right: each fixed input bundle with $F(x) \geq q$ gives a line in w_1 ; $c(\cdot, q)$ is their lower envelope, hence concave, with slope $x_1(w, q)$.

Auction payment formula

Envelope characterization

In a single-object auction, type v wins with probability $q(v)$ and pays $p(v)$, so $U(v) = vq(v) - p(v)$.

When the direct mechanism is incentive compatible, U is absolutely continuous and

$$U'(v) = q(v) \text{ a.e.}, \quad U(v) = U(\underline{v}) + \int_{\underline{v}}^v q(t) dt.$$

Since $U(v) = vq(v) - p(v)$,

$$p(v) = vq(v) - U(\underline{v}) - \int_{\underline{v}}^v q(t) dt.$$

Once the allocation rule is fixed, payments are pinned down up to the lowest type's surplus. With $q(v) = v$ on $[0, 1]$ and $U(0) = 0$: $U(v) = \int_0^v t dt = v^2/2$ and $p(v) = v \cdot v - v^2/2 = v^2/2$.

Further reading: [Roughgarden, auction notes](#).

Payments from an allocation rule

Two allocation rules

Types are distributed on $[0, 1]$ and $U(0) = 0$.

- (a) For $q(v) = v^2$, find $U(v)$ and $p(v)$, and check $U'(v) = q(v)$.
- (b) For the reserve rule $q(v) = \mathbf{1}\{v \geq r\}$ with $r \in (0, 1)$, find $U(v)$ and $p(v)$ for $v \geq r$. What does a winner pay?
- (c) In (b), where does U fail to be differentiable, and why does the envelope formula still determine payments?

Payments from an allocation rule

Under the reserve rule the winner pays the reserve

(a) $U(v) = \int_0^v t^2 dt = v^3/3$, and $p(v) = vq(v) - U(v) = v^3 - v^3/3 = \frac{2}{3}v^3$. Differentiating, $U'(v) = v^2 = q(v)$.

(b) For $v \geq r$, $U(v) = \int_0^v \mathbf{1}\{t \geq r\} dt = v - r$, so

$$p(v) = v \cdot 1 - (v - r) = r.$$

A winner pays the reserve. For $v < r$ the type neither wins nor pays: $q = p = U = 0$.

(c) U has a kink at $v = r$, where q jumps from 0 to 1; the left and right derivatives are 0 and 1. The integral form $U(v) = U(0) + \int_0^v q(t) dt$ still holds at every type, since U is the integral of a bounded q , so payments are pinned down at the kink as well.

Section 7

Worked example

Cobb-Douglas utility maximization

Let $\alpha, \beta, p_1, p_2, w > 0$. Maximize $x^\alpha y^\beta$ subject to $p_1 x + p_2 y \leq w$ and $x, y > 0$. Monotonicity makes the budget bind. By the log trick, maximize $\alpha \ln x + \beta \ln y$ instead, with

$$\mathcal{L} = \alpha \ln x + \beta \ln y + \lambda(w - p_1 x - p_2 y).$$

The first-order conditions are

$$\frac{\alpha}{x} = \lambda p_1, \quad \frac{\beta}{y} = \lambda p_2, \quad p_1 x + p_2 y = w.$$

Thus

$$p_1 x = \frac{\alpha}{\lambda}, \quad p_2 y = \frac{\beta}{\lambda}.$$

Video: [Economics in Many Lessons, Utility Maximization with a Cobb-Douglas Utility Function.](#)

Cobb-Douglas utility maximization: solve the conditions

Adding the two expenditure equations and using the binding budget gives

$$w = p_1x + p_2y = \frac{\alpha + \beta}{\lambda} \Rightarrow \lambda^* = \frac{\alpha + \beta}{w}.$$

Substitution into the first-order conditions yields

$$x^* = \frac{\alpha}{\alpha + \beta} \frac{w}{p_1}, \quad y^* = \frac{\beta}{\alpha + \beta} \frac{w}{p_2}.$$

This one problem reuses three earlier chapters: Functions gives the strictly increasing log transformation, Differentiation gives the FOCs, and Convexity gives strict concavity of the log objective and convexity of the budget set. Hence the FOCs identify the **unique global maximizer**.

Further reading: [Border, Preference and Demand Examples, Sections 1–2.](#)

Expenditure shares and the multiplier

Expenditure shares

Demands spend fixed income shares $\alpha/(\alpha + \beta)$ and $\beta/(\alpha + \beta)$ on the two goods.

Multiplier

$$\lambda^* = \frac{\alpha + \beta}{w} = \tilde{v}_w.$$

It is the envelope derivative of the indirect log-value.

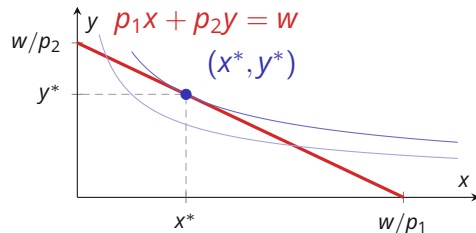


Figure 33 · Budget line and the tangent Cobb-Douglas indifference curve. With $\alpha : \beta = 1 : 2$, x^* is one third of w/p_1 and y^* two thirds of w/p_2 .

Cobb-Douglas utility with an upper bound on one good

Let $\alpha, \beta, p_x, p_y, w > 0$ and $0 < \bar{x} < w/p_x$. Solve

$$\max_{x,y>0} x^\alpha y^\beta \quad \text{s.t.} \quad p_x x + p_y y \leq w, \quad x \leq \bar{x}.$$

Use the log objective.

- Determine when the upper bound binds, find (x^*, y^*) in each case, and compute the multipliers on the budget and on the upper bound.
- Treat $\bar{x}$ as a parameter. Describe $x^*(\bar{x})$, $y^*(\bar{x})$ and $\mu^*(\bar{x})$ on both sides of the threshold $\bar{x} = x^u$, write the value $V(\bar{x})$ of the log objective on each side, and check $V'(\bar{x}) = \mu^*(\bar{x})$.

Cobb-Douglas utility with an upper bound: choices

Without the upper bound, the demand for x is

$$x^u = \frac{\alpha}{\alpha + \beta} \frac{w}{p_x}.$$

Monotonicity makes the budget bind, so the choice is over x alone with a concave objective peaking at x^u . Cutting that interval at $\bar{x}$ therefore gives

$$x^* = \min\{x^u, \bar{x}\}, \quad y^* = \frac{w - p_x x^*}{p_y}.$$

If $\bar{x} > x^u$, the upper bound is slack. If $\bar{x} < x^u$, it binds and $(x^*, y^*) = (\bar{x}, (w - p_x \bar{x})/p_y)$. At $\bar{x} = x^u$, it binds with a zero multiplier. Strict concavity gives uniqueness and globality.

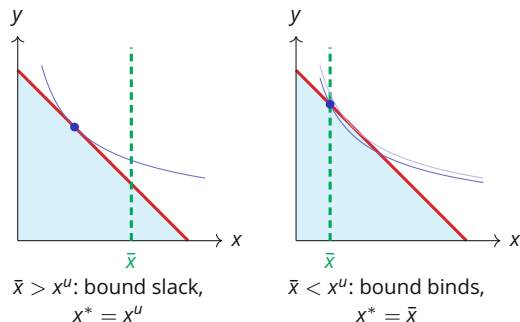


Figure 34 · The cap $x \leq \bar{x}$ (dashed) truncates the budget set. Left: $\bar{x}$ above the unconstrained demand x^u , the bound is slack. Right: $\bar{x}$ below x^u , the optimum lies on the cap where the indifference curve crosses the budget line.

Cobb-Douglas utility with an upper bound: multipliers

With $\mathcal{L} = \alpha \ln x + \beta \ln y + \lambda(w - p_x x - p_y y) + \mu(\bar{x} - x)$, the conditions are

$$\frac{\alpha}{x} - \lambda p_x - \mu = 0, \quad \frac{\beta}{y} - \lambda p_y = 0, \quad \lambda(w - p_x x - p_y y) = \mu(\bar{x} - x) = 0,$$

and monotonicity makes the budget bind, $p_x x + p_y y = w$.

Cap slack. Then $\mu = 0$, so $x = \alpha/(\lambda p_x)$ and $y = \beta/(\lambda p_y)$; the budget gives $(\alpha + \beta)/\lambda = w$, hence $\lambda^* = (\alpha + \beta)/w$ and $x^* = x^u$.

Cap binding. Then $x = \bar{x}$ and $y = (w - p_x \bar{x})/p_y$, so the y equation gives $\lambda^* = \beta/(w - p_x \bar{x})$ and the x equation gives

$$\mu^* = \frac{\alpha}{\bar{x}} - \lambda^* p_x = \frac{\alpha}{\bar{x}} - \frac{\beta p_x}{w - p_x \bar{x}}.$$

Both denominators are positive, so $\mu^* \geq 0$ is equivalent to $\alpha(w - p_x \bar{x}) \geq \beta p_x \bar{x}$, that is $\alpha w \geq (\alpha + \beta)p_x \bar{x}$, that is $\bar{x} \leq x^u$. The sign condition on the multiplier therefore holds on the same parameters where the cap binds. The two multipliers are the shadow prices of wealth and of the cap for the optimized log value.

Cobb-Douglas utility with an upper bound: the cap as a parameter

(b) Above the threshold, $\bar{x} \geq x^u$: the cap is slack,

$$x^* = x^u = \frac{\alpha}{\alpha + \beta} \frac{w}{p_x}, \quad y^* = \frac{\beta}{\alpha + \beta} \frac{w}{p_y}, \quad \mu^* = 0,$$

all free of $\bar{x}$. The value $V(\bar{x}) = \alpha \ln x^u + \beta \ln y^*$ is constant, so $V'(\bar{x}) = 0 = \mu^*$.

Below the threshold, $\bar{x} < x^u$: the cap binds, $x^* = \bar{x}$ and $y^* = (w - p_x \bar{x})/p_y$, so

$$V(\bar{x}) = \alpha \ln \bar{x} + \beta \ln \frac{w - p_x \bar{x}}{p_y}, \quad V'(\bar{x}) = \frac{\alpha}{\bar{x}} - \frac{\beta p_x}{w - p_x \bar{x}} = \mu^*(\bar{x}).$$

Across the threshold. As $\bar{x} \uparrow x^u$, $\mu^*(\bar{x}) \rightarrow 0$, so both left and right derivatives of V agree at x^u and V is differentiable there. The optimizer $x^*(\bar{x}) = \min\{x^u, \bar{x}\}$ is continuous with a kink at x^u , and the active set changes there: the cap enters the binding set as $\bar{x}$ falls below x^u .

Section 8

Further reading

Optimization references

Recommended references

Osborne, *Mathematical Methods for Economic Theory*, optimization sections [\[link\]](#).

Echenique, *Constrained Maximization* [\[link\]](#).

Border, *Miscellaneous Notes on Optimization Theory and Related Topics* [\[link\]](#).

Comparative statics and extensions

Selected references

Echenique, *Monotone Comparative Statics* [link];

Border, *Envelope Theorem* [link].

Border, *Single Crossing Properties* [link] and *Lattices and Supermodularity* [link].

Sarver, *Microeconomic Theory Lecture Notes*, Chapters 1–2 [link].

Original sources: Milgrom and Shannon (1994) [link]; Topkis (1998) [link].